

On the spectrum of minimum ℓ -degree Turán densities

Heng Li*

Xizhi Liu†

August 26, 2026

Abstract

For integers $r > \ell \geq 2$, the minimum ℓ -degree of an r -graph is the smallest number of edges containing any one of its ℓ -sets. Given a family $\mathcal{F}$ of r -graphs, its ℓ -degree Turán density $\pi_\ell^r(\mathcal{F})$ is the asymptotically largest normalized minimum ℓ -degree of an $\mathcal{F}$ -free r -graph. We write $\Pi_{\ell,\infty}^r$ for the set of these densities over all finite or infinite forbidden families whose members contain an edge, and we also allow the empty family. We prove that $\Pi_{\ell,\infty}^r = [0, 1]$ whenever $\ell \geq 3$, or $\ell = 2$ and $r \geq 4$, leaving the three-uniform codegree case $(r, \ell) = (3, 2)$ as the only remaining case. In particular, this answers a question of Mubayi and Zhao about the codegree Turán spectrum for every $r \geq 4$.

2020 Mathematics Subject Classification. 05C65, 05C35, 05D05.

Keywords. Hypergraph, minimum ℓ -degree, codegree, Turán density, infinite forbidden family.

1 Introduction

For an r -graph G and an ℓ -set $S \subseteq V(G)$, the ℓ -degree of S is the number of edges containing it. The $(r - 1)$ -degree is usually called the codegree. The minimum ℓ -degree Turán problem asks for the largest possible minimum ℓ -degree among r -graphs avoiding a prescribed family.

It is well known that the cases $\ell = 0$ and $\ell = 1$ reduce to ordinary Turán density, so we restrict attention to $\ell \geq 2$. Keevash [17] surveys the classical hypergraph Turán problem and its main methods. The set of attainable ordinary Turán densities is already highly nontrivial. Frankl and Rödl [11] disproved the jump conjecture of Erdős [5], and Pikhurko [27] proved that the ordinary Turán density spectrum over all forbidden families has the cardinality of the continuum and that finite families can have irrational density. Liu and Pikhurko [20, Theorem 1.1] subsequently found a terminal interval in this spectrum for every uniformity at least three, answering questions of Frankl, Peng, Rödl, and Talbot [10, Question 6.1] and Grosu [13, Problem 5]. Related interval phenomena also occur in uniformly dense Turán problems: Li, Liu, and Pikhurko [19, Theorem 1.1 and Corollary 6.1] proved that the $(r - 2)$ -uniform Turán spectra of r -graphs, taken over all forbidden families, contain a terminal interval for all $r \geq 3$.

Turning to minimum ℓ -degrees, Mubayi and Zhao [25, Theorem 1.6] proved that, for every $r \geq 3$, the codegree Turán spectrum over all forbidden families is dense in $[0, 1]$. More precisely, for every $r \geq 3$, every $\alpha \in [0, 1)$, and every $\varepsilon > 0$, there is a finite family of r -graphs whose codegree Turán density lies strictly between α and $\alpha + \varepsilon$. They further conjectured [25, Conjecture 1.7] that every $\alpha \in [0, 1)$ is realized exactly by an infinite family $\mathcal{F}$ for which every finite subfamily has codegree Turán density strictly larger than α . Lo and Markström [21, Theorem 1.7] proved that the corresponding minimum ℓ -degree spectrum is dense in $[0, 1]$ for every $r > \ell > 1$. More recently, Gao, Pikhurko, Rong, and Sun [12, Theorem 1.1] proved that, for every $r \geq 3$, every rational number in $[0, 1)$ is the codegree Turán density of a finite family of r -graphs. We prove that this spectrum is the full interval in all but one of the cases considered by Lo and Markström.

The codegree Turán problem for a single forbidden hypergraph has developed along a different line. Mubayi [24] determined the codegree Turán density of the Fano plane, and Keevash [16] subsequently determined its exact codegree threshold for all sufficiently large orders. Keevash and Zhao [18] studied projective geometries more generally; Fang and Chen [9] recently resolved their positivity question and

*School of Mathematics, Shandong University, Jinan, China, and Extremal Combinatorics and Probability Group, Institute for Basic Science, Daejeon, South Korea. Email: heng.li@sdu.edu.cn.

†School of Mathematical Sciences, University of Science and Technology of China, Hefei, Anhui 230026, China. Email: liuxizhi@ustc.edu.cn.

obtained further exact codegree Turán densities. Other representative results include the asymptotic bounds on the codegree Turán densities of complete hypergraphs due to Lo and Zhao [22] and the determination of the codegree threshold of K_4^- by Falgas-Ravry, Pikhurko, Vaughan, and Volec [7].

At the lower end of the single-forbidden spectrum, Piga and Schülke [26, Theorem 1.1] constructed r -graphs whose codegree Turán densities are arbitrarily small and positive, while Ding, Lamaison, Liu, Wang, and Yang [4] studied 3-graphs with vanishing codegree Turán density. For every $r \geq 3$, Ai, Ding, Liu, and Yang [2] proved that zero is an accumulation point of the single-forbidden minimum 2-degree Turán spectrum, and Fang and Chen [8] constructed linear r -graphs with arbitrarily small positive codegree Turán density. The ordinary single-forbidden spectrum likewise has an accumulation point in every uniformity at least three [3, Theorem 1.1]. These codegree results describe individual values and behavior near zero in single-forbidden spectra; our results instead classify the spectrum when arbitrary forbidden families are allowed.

For an arbitrary, possibly infinite family $\mathcal{F}$ of r -graphs, each containing an edge, let $\pi_\ell^r(\mathcal{F})$ be its normalized minimum ℓ -degree Turán density, and let $\Pi_{\ell,\infty}^r$ be the set of all such densities. The empty family is allowed. Precise definitions are given in Section 2. Our main result determines this spectrum in every case with $\ell \geq 2$ apart from the three-uniform codegree case.

Theorem 1.1. *For all integers $r > \ell \geq 2$ with $(r, \ell) \neq (3, 2)$, one has $\Pi_{\ell,\infty}^r = [0, 1]$.*

We prove the cases $\ell \geq 3$, $(r, \ell) = (4, 2)$, and $r \geq 5$ with $\ell = 2$ in Sections 3 to 5, respectively.

The only remaining question is whether $\Pi_{2,\infty}^3 = [0, 1]$. Our result establishes the exact-realization conclusion of [25, Conjecture 1.7] for codegrees in every uniformity $r \geq 4$, but does not show that each finite subfamily can be required to have density strictly larger than the target.

Our proof divides naturally according to whether $\ell \geq 3$ or $\ell = 2$. Several of its constructions develop the cyclic part method of Mubayi and Zhao [25, Definition 3.3 and Sections 3.2–3.3], whose underlying idea is to organize the vertex classes cyclically and control the construction through prescribed links. The transversal-link construction extends this principle to general ℓ and permits the link classes to vary. The same idea reappears in the minimum 2-degree argument, with cyclically arranged parts for 4-graphs and symmetric parts carrying ordinary 3-graph links in higher uniformities.

For $\ell \geq 3$, we combine the terminal interval in the spectrum of ordinary hypergraph Turán densities [20, Theorem 1.1] with the transversal-link construction. This construction multiplies an ordinary Turán density by a factor from a dense set. For any prescribed real number in $(0, 1)$, we choose a slightly larger factor so that the quotient of this number by the factor lies in the terminal interval, thereby realizing the number as a density. The classes of edgeless and of all r -graphs realize the endpoints 0 and 1, respectively. In particular, taking $\ell = r - 1$ gives $\Pi_{r-1,\infty}^r = [0, 1]$ for every $r \geq 4$ and therefore determines the codegree Turán spectrum over all forbidden families in every uniformity at least four.

For $\ell = 2$, this approach cannot start from the ordinary graph Turán spectrum, which, by the seminal Erdős–Stone theorem [6], contains no nondegenerate interval. We first construct an interval adjacent to zero for 4-graphs and then expand it to $[0, 1]$ through a sequence of continuous transfer constructions with overlapping ranges. For larger uniformities, one construction lifts the low-density interval from the four-uniform case, while constructions based on ordinary 3-graph links cover densities close to one. These ranges overlap uniformly for $r \geq 8$; a sharper form of the lift handles $r = 7$, and two-part constructions bridge the remaining gaps for $r = 5, 6$.

The paper is organized as follows. Section 2 introduces minimum ℓ -degree Turán densities and admissible monotone classes, and Section 3 proves Theorem 1.1 for $\ell \geq 3$. The four-uniform minimum 2-degree case is proved in Section 4, and Section 5 treats minimum 2-degrees in every larger uniformity. Finally, Section 6 records the exact computational certificates and their verification protocol.

2 Preliminaries

This section records the definitions and general tools used throughout this work. We first introduce minimum ℓ -degree Turán densities and the corresponding spectra over all forbidden families, then recast them in terms of admissible monotone classes, record the terminal-interval theorem for ordinary Turán densities, and finally prove a vertex-deletion lemma for near-extremal ordinary hypergraphs.

An r -graph is a finite simple r -uniform hypergraph. We identify a hypergraph with its edge set, and hence write $|H|$ for the number of edges of H . We write $V(G)$ for the vertex set of a hypergraph G and $v(G) := |V(G)|$ for its order. For integers x and $p \geq 1$, write $(x)_p := x(x-1) \cdots (x-p+1)$. Unless stated otherwise, the parts in every partition considered below are allowed to be empty.

For disjoint vertex classes $U_1, \dots, U_t$ and positive integers $c_1, \dots, c_t$, the *intersection profile* $U_1^{c_1} \cdots U_t^{c_t}$ consists of the $(\sum_j c_j)$ -sets meeting U_j in exactly c_j vertices for every j ; exponents equal to one are omitted. When such a profile is allowed, an arbitrary subfamily of its sets may be used.

For an r -graph G and a set $S \subseteq V(G)$ with $|S| < r$, the *link* of S in G is the $(r - |S|)$ -graph on $V(G) \setminus S$ with edge set $\{e \setminus S : e \in G \text{ and } S \subseteq e\}$. For $0 \leq \ell < r$ and $v(G) \geq \ell$, define the degree of an ℓ -set $S \subseteq V(G)$ and the minimum ℓ -degree of G by

$$d_G(S) := |\{e \in G : S \subseteq e\}|, \quad \text{and} \quad \delta_\ell(G) := \min_{S \in \binom{V(G)}{\ell}} d_G(S).$$

Thus $\delta_0(G) = |G|$ and $\delta_{r-1}(G)$ is the minimum codegree.

For a family $\mathcal{F}$ of r -graphs, let $\text{Forb}(\mathcal{F})$ denote the class of r -graphs containing no member of $\mathcal{F}$ as a (not necessarily induced) subgraph. For $n \geq r$, write $\text{ex}_\ell(n, \mathcal{F})$ for the maximum of $\delta_\ell(G)$ over all n -vertex r -graphs in $\text{Forb}(\mathcal{F})$. The normalized minimum ℓ -degree Turán density is

$$\pi_\ell^r(\mathcal{F}) := \lim_{n \rightarrow \infty} \frac{\text{ex}_\ell(n, \mathcal{F})}{\binom{n-\ell}{r-\ell}}.$$

Lo and Markström [21, Proposition 1.5] proved that $\text{ex}_\ell(n, \mathcal{F}) / \binom{n}{r-\ell}$ converges for every forbidden family $\mathcal{F}$, finite or infinite. Since $\binom{n}{r-\ell} / \binom{n-\ell}{r-\ell} \rightarrow 1$, the limit in the preceding definition exists.

Define the spectrum of minimum ℓ -degree Turán densities over all forbidden families by

$$\Pi_{\ell, \infty}^r := \{\pi_\ell^r(\mathcal{F}) : \mathcal{F} \text{ is an arbitrary, possibly infinite family of } r\text{-graphs, each with an edge}\}.$$

The empty forbidden family is allowed and satisfies $\pi_\ell^r(\emptyset) = 1$. If a forbidden family $\mathcal{F}$ contains an edgeless r -graph on f vertices, then $\text{Forb}(\mathcal{F})$ has no member of order at least f ; we exclude this degenerate situation.

A class $\mathcal{P}$ of r -graphs is *monotone* if it is invariant under isomorphism and closed under deleting vertices and edges. It is *admissible* if it contains the edgeless r -graph of every order. Monotonicity gives the identity $\mathcal{P} = \text{Forb}(\{F : F \text{ is an } r\text{-graph and } F \notin \mathcal{P}\})$. Conversely, $\text{Forb}(\mathcal{F})$ is monotone for every forbidden family $\mathcal{F}$, and it is admissible when every member of $\mathcal{F}$ has an edge. We shall therefore construct admissible monotone classes and pass back to forbidden families only at the end.

For an admissible monotone class $\mathcal{P}$ of r -graphs, define

$$\text{dens}_\ell^r(\mathcal{P}) := \lim_{n \rightarrow \infty} \frac{\max\{\delta_\ell(G) : G \in \mathcal{P} \text{ and } v(G) = n\}}{\binom{n-\ell}{r-\ell}}.$$

By the identity $\mathcal{P} = \text{Forb}(\{F : F \notin \mathcal{P}\})$, the same result of Lo and Markström and the same normalization conversion show that this limit exists. We also use the ordinary Turán density of an admissible monotone class $\mathcal{Q}$ of s -graphs, defined by

$$\text{dens}(\mathcal{Q}) := \lim_{n \rightarrow \infty} \frac{\max\{|H| : H \in \mathcal{Q} \text{ and } v(H) = n\}}{\binom{n}{s}}.$$

The ordinary limit follows from the standard Katona–Nemetz–Simonovits averaging argument [15].

The following theorem supplies the interval of ordinary Turán densities used in the later constructions.

Theorem 2.1 ([20, Theorem 1.1]). *For every integer $s \geq 3$, there is a constant $c_s \in (0, 1)$ such that, for every $\beta \in [c_s, 1]$, there is an admissible monotone class $\mathcal{Q}$ of s -graphs with $\text{dens}(\mathcal{Q}) = \beta$.*

Every proper admissible monotone class $\mathcal{P}$ of r -graphs satisfies $\text{dens}_\ell^r(\mathcal{P}) < 1$ for $0 \leq \ell < r$. Indeed, if $F \notin \mathcal{P}$ and $f = v(F)$, then admissibility ensures that F has an edge, while monotonicity makes every member of $\mathcal{P}$ $K_f^{(r)}$ -free. Double counting missing edges in f -vertex sets and then averaging the ℓ -degrees gives $\delta_\ell(G) / \binom{n-\ell}{r-\ell} \leq |G| / \binom{n}{r} \leq 1 - 1 / \binom{f}{r}$ for every sufficiently large $G \in \mathcal{P}$ of order n .

For each class constructed below, we verify monotonicity and admissibility. Once such a class has been shown to have density $\alpha < 1$, it is proper. Its complement is then a nonempty family of r -graphs, each with at least one edge. We shall use this passage from classes to the forbidden families in the definition of $\Pi_{\ell,\infty}^r$ without further comment. The class of all r -graphs corresponds to the empty forbidden family and realizes the remaining endpoint 1.

The final ingredient in this section is the following vertex-deletion lemma, which produces near-extremal s -graphs with large minimum vertex degree.

Lemma 2.2. *Let $s \geq 2$, and let $\mathcal{Q}$ be an admissible monotone class of s -graphs with ordinary Turán density β . For every $\varepsilon > 0$, there are arbitrarily large $H \in \mathcal{Q}$ such that*

$$|H| \geq (\beta - \varepsilon) \binom{v(H)}{s}, \quad \text{and} \quad \delta_1(H) \geq (\beta - 2\varepsilon) \binom{v(H) - 1}{s - 1}.$$

Proof. If $\beta = 0$, take edgeless s -graphs of arbitrary order. Suppose that $\beta > 0$ and set $\eta := \min\{\varepsilon/2, \beta/4\}$. Choose $c \in (0, 1)$ so that $(1 - \beta + 2\eta)c^s < \eta$. For every sufficiently large N , choose an N -vertex $G \in \mathcal{Q}$ with $|G| \geq (\beta - \eta) \binom{N}{s}$.

While the current s -graph has order m , delete a vertex of degree smaller than $(\beta - 2\eta) \binom{m-1}{s-1}$, if one exists. If the process reaches order m , telescoping gives $|G_m| \geq (\beta - 2\eta) \binom{m}{s} + \eta \binom{N}{s}$. If $m \leq cN$, then $\binom{m}{s} \leq c^s \binom{N}{s}$, so the right-hand side exceeds $\binom{m}{s}$, which is impossible. Thus the process stops with $m > cN$.

At that point its minimum vertex degree is at least $(\beta - 2\eta) \binom{m-1}{s-1}$, while the displayed edge bound and $m \leq N$ give edge density at least $\beta - \eta$. Since $\eta \leq \varepsilon/2$, both claimed bounds follow, and $m \rightarrow \infty$ as $N \rightarrow \infty$ through sufficiently large integers. $\square$

3 Minimum ℓ -degree for $\ell \geq 3$

We prove the case $\ell \geq 3$ by a transversal-link construction adapted from the cyclic part-and-window mechanism of Mubayi and Zhao [25, Definition 3.3 and Section 3.3]. In their codegree construction, an $(r - 1)$ -set contained in one part is completed by vertices in a prescribed forward window, while all r -sets meeting each part in at most $r - 2$ vertices form the baseline. We replace the completing vertex by a multi-part transversal and allow each resulting link to belong to an arbitrary monotone class. The resulting construction, which will also be used later for minimum 2-degrees, converts ordinary Turán densities of ℓ -graphs into minimum ℓ -degree Turán densities of r -graphs.

Fix integers $r > \ell \geq 2$ and $p := r - \ell$. Choose integers a, b with $p \leq b \leq a - \ell$, and let $\mathcal{Q}$ be an admissible monotone class of ℓ -graphs with $\text{dens}(\mathcal{Q}) = \beta$. All part indices in this section are read modulo a .

Given a partition $V = V_0 \dot{\cup} \dots \dot{\cup} V_{a-1}$, whose parts may be empty, call a p -set T an i -transversal if it meets p distinct parts V_j with $j \in \{i + 1, \dots, i + b\}$, once in each part.

Construction 3.1. *Let $\mathcal{W}_{a,b}^{r,\ell}(\mathcal{Q})$ be the class of all r -graphs G admitting a partition $V(G) = V_0 \dot{\cup} \dots \dot{\cup} V_{a-1}$ with the following properties.*

- (W1) *For every $i \in \mathbb{Z}_a$ and every i -transversal T , the ℓ -graph $L_{i,T} := \{S \in \binom{V_i}{\ell} : S \cup T \in G\}$ belongs to $\mathcal{Q}$.*
- (W2) *Every edge of G is either a link-type edge $S \cup T$, where $S \in L_{i,T}$ for some i and some i -transversal T , or an edge meeting each part in at most $\ell - 1$ vertices.*

The link-type edges realize the prescribed $\mathcal{Q}$ -links, whereas the second edge type provides completions for ℓ -sets not contained in a single part.

Theorem 3.2. *Let $r > \ell \geq 2$, put $p := r - \ell$, and let a, b be integers satisfying $p \leq b \leq a - \ell$. If $\mathcal{Q}$ is an admissible monotone class of ℓ -graphs with $\text{dens}(\mathcal{Q}) = \beta$, then $\mathcal{W}_{a,b}^{r,\ell}(\mathcal{Q})$ is an admissible monotone class with $\text{dens}_\ell^r(\mathcal{W}_{a,b}^{r,\ell}(\mathcal{Q})) = \beta(b)_p / a^p$.*

Proof. We first verify that the resulting class is admissible and monotone. A link-type edge has ℓ vertices in a unique part V_i and at most one vertex in every other part, so the two edge types are disjoint. Deleting a link-type edge deletes one edge from the corresponding link, while deleting a vertex restricts the affected

links and removes the transversals that contain it. Thus both operations preserve the conditions in [Construction 3.1](#), because $\mathcal{Q}$ is monotone. Since $\mathcal{Q}$ is admissible, every empty link belongs to $\mathcal{Q}$, and the resulting class contains the edgeless r -graph of every order.

For this calculation, put $\lambda := (b)_p/a^p = p!(\frac{b}{p})/a^p$.

For the lower bound, partition an n -vertex set as evenly as possible into $V_0, \dots, V_{a-1}$, and let $\mathcal{T}_i$ be the set of i -transversal p -sets. Then

$$|\mathcal{T}_i| = \left(\binom{b}{p} \frac{1}{a^p} + o(1) \right) n^p = (\lambda + o(1)) \binom{n-\ell}{p}.$$

By the convergence in the definition of $\text{dens}(\mathcal{Q})$, we may choose a sequence $\tau_n \rightarrow 0$ sufficiently slowly that $\tau_n^2 n^p \gg \log n$ and, for every i , an ℓ -graph $H_i \in \mathcal{Q}$ on V_i with edge density at least $\beta - \tau_n$. For every $T \in \mathcal{T}_i$, take an independent uniformly random relabeling of H_i as $L_{i,T}$. For fixed $S \in \binom{V_i}{\ell}$, the number of $T \in \mathcal{T}_i$ with $S \in L_{i,T}$ has expectation at least $(\beta - \tau_n)|\mathcal{T}_i|$. Since $|\mathcal{T}_i| = \Theta(n^p)$, Hoeffding's inequality [\[14\]](#) and a union bound over all $O(n^\ell)$ choices of S show that the relabelings can be chosen so that every such S belongs to at least $(\beta - 2\tau_n)|\mathcal{T}_i|$ of the links. Include all resulting link-type edges and all r -sets meeting each part in at most $\ell - 1$ vertices.

If $S \subseteq V_i$, then $d_G(S) \geq (\beta\lambda - o(1))\binom{n-\ell}{p}$. Otherwise S meets at most ℓ parts, so choosing one vertex from each of p further parts produces at least

$$\binom{a-\ell}{p} \left(\frac{n}{a} - O(1) \right)^p = \left(\frac{(a-\ell)_p}{a^p} - o(1) \right) \binom{n-\ell}{p}$$

completions by edges of the second type. Since $(a-\ell)_p \geq (b)_p \geq \beta(b)_p$, every ℓ -set has the required normalized degree.

For the upper bound, fix $\varepsilon > 0$ and choose $N \geq \ell$ so that $|H| \leq (\beta + \varepsilon)\binom{v(H)}{\ell}$ whenever $H \in \mathcal{Q}$ and $v(H) \geq N$. Let G be an n -vertex member of $\mathcal{W}_{a,b}^{r,\ell}(\mathcal{Q})$ with a witnessing partition whose part sizes are $w_0, \dots, w_{a-1}$, and let $\mathcal{T}_i$ denote the set of i -transversal p -sets for this partition. Set $z_i = w_i$ when $w_i \geq N$, and $z_i = 0$ otherwise. For all sufficiently large n , some z_i is positive. We need the following weighted averaging claim over the indices with $z_i > 0$.

Claim 3.3. *Some i with $z_i > 0$ satisfies $\sum_{j=1}^b z_{i+j} \leq \frac{b}{a} \sum_{j=0}^{a-1} z_j$.*

Proof. Let $J := \{i: z_i > 0\}$, listed in cyclic order, and for $i \in J$ let g_i be the positive cyclic distance from i to the next member of J . The corresponding cyclic arcs partition $\mathbb{Z}_a$, so $\sum_{i \in J} g_i = a$. For fixed $j \in J$, the indices $i \in J$ for which $j \in (i, i+b]$ form a consecutive block immediately preceding j . The arcs associated with this block fit inside the b positions preceding j and therefore have total length at most b . Thus the coefficient of z_j in the weighted average below is at most b/a , and hence

$$\sum_{i \in J} \frac{g_i}{a} \sum_{t=1}^b z_{i+t} \leq \frac{b}{a} \sum_{j \in J} z_j.$$

The weights g_i/a sum to one, so some $i \in J$ satisfies the claim. □

By the claim and $\sum_j z_j \leq n$, choose i with $w_i \geq N$ and $\sum_{j=1}^b z_{i+j} \leq bn/a$. For each $j \in \{1, \dots, b\}$ with $z_{i+j} = 0$, we have $w_{i+j} < N$, so restoring the omitted part sizes adds less than bN . Hence $\sum_{j=1}^b w_{i+j} \leq bn/a + bN$.

No edge of the second type contains an ℓ -set in V_i , and every link-type edge containing such a set arises from a unique link $L_{i,T}$. Therefore double counting gives the exact identity

$$\frac{1}{\binom{w_i}{\ell}} \sum_{S \in \binom{V_i}{\ell}} d_G(S) = \frac{1}{\binom{w_i}{\ell}} \sum_{T \in \mathcal{T}_i} |L_{i,T}| \leq (\beta + \varepsilon)|\mathcal{T}_i|.$$

Some $S \in \binom{V_i}{\ell}$ consequently has degree at most $(\beta + \varepsilon)|\mathcal{T}_i|$. Maclaurin's inequality now yields

$$|\mathcal{T}_i| \leq \binom{b}{p} \left(\frac{1}{b} \sum_{j=1}^b w_{i+j} \right)^p \leq \binom{b}{p} \left(\frac{n}{a} + N \right)^p.$$

After normalization and passage to the limsup, the density is at most $(\beta + \varepsilon)(b)_p/a^p$. Letting $\varepsilon \rightarrow 0$ proves the theorem. $\square$

We now combine the transversal-link construction with [Theorem 2.1](#) to prove the $\ell \geq 3$ part of [Theorem 1.1](#).

Proof of [Theorem 1.1](#) for $\ell \geq 3$. The empty forbidden family realizes the value 1, and the class of edgeless r -graphs realizes the value 0. Fix $\alpha \in (0, 1)$, put $p := r - \ell$, and let $c_\ell < 1$ be supplied by [Theorem 2.1](#). Choose $x \in (\alpha, \min\{\alpha/c_\ell, 1\})$. For every sufficiently large integer a , set $b := \lfloor x^{1/p}a \rfloor$. Since $(1 - x^{1/p})a \rightarrow \infty$, the inequalities $p \leq b \leq a - \ell$ hold for all sufficiently large a . Moreover, the p consecutive factors in $(b)_p/a^p$ all tend to $x^{1/p}$, and hence $(b)_p/a^p \rightarrow x$. We may therefore choose a so that $\alpha < \lambda := (b)_p/a^p < \min\{\alpha/c_\ell, 1\}$. Set $\beta := \alpha/\lambda$. Then $c_\ell < \beta < 1$, so [Theorem 2.1](#) gives an admissible monotone class $\mathcal{Q}$ of ℓ -graphs satisfying $\text{dens}(\mathcal{Q}) = \beta$. By [Theorem 3.2](#), the class $\mathcal{W}_{a,b}^{r,\ell}(\mathcal{Q})$ has density $\beta\lambda = \alpha$. The passage described at the end of [Section 2](#) now yields a forbidden family in which every member has an edge. $\square$

The argument genuinely changes at $\ell = 2$. In that case $\mathcal{Q}$ is a monotone class of ordinary graphs. The Erdős–Stone theorem [6] implies that the spectrum over admissible monotone graph classes is $\{0, 1/2, 2/3, 3/4, \dots, 1\}$, which contains no nondegenerate interval. Although the formula in [Theorem 3.2](#) remains valid, the possible values of β contain no interval.

4 The four-uniform minimum 2-degree case

Throughout this section, all hypergraphs are 4-uniform and the degree parameter is $\ell = 2$. We abbreviate $\text{dens}_2^4(\mathcal{P})$ to $\text{dens}_2(\mathcal{P})$.

Theorem 4.1. $\Pi_{2,\infty}^4 = [0, 1]$.

We prove the theorem in four stages. First, the satellite construction and ordinary 3-graph vertex links give a nondegenerate interval adjacent to zero. Then [Proposition 4.6](#) extends the realized interval to $3/38$, the D_{25} construction in [Proposition 4.9](#) extends it to $2/25$, and the high-density transfers complete the interval $[0, 1]$. Each stage preserves admissibility and monotonicity, accounts for partitions with sublinear parts, verifies the degree bounds for pairs meeting distinct parts, and has an explicit continuous transfer function. The finitely many rational choices needed for some extensions are supplied as exact computational certificates available at https://xliu2022.github.io/CodegreeTuranInterval_certificates/, while the required analytic inequalities are proved throughout the stated parameter ranges.

4.1 The satellite construction and an initial interval

We begin with a heterogeneous satellite construction whose density map will be used repeatedly to enlarge realized intervals. Its cyclic organization of satellite parts and local links is motivated by the same part-and-window principle [25, Sections 3.2–3.3], although the central part and graph-valued links below are tailored to minimum 2-degrees rather than being a specialization of that construction. Let $m \geq 3$ be an integer, and let H be a triangle-free graph on $\mathbb{Z}_m$ satisfying $1 \leq \delta(H) \leq \Delta(H) \leq 2$. In the partition below, A is the *central part* and $B_0, \dots, B_{m-1}$ are the *satellite parts*; the satellite indices are read modulo m . For each $i \in \mathbb{Z}_m$, choose $s_i \in \{2, 3, \dots\} \cup \{\infty\}$ and put

$$\gamma_i := \begin{cases} 1 - 1/s_i, & s_i < \infty, \\ 1, & s_i = \infty. \end{cases}$$

Write $\gamma := (\gamma_i)_{i \in \mathbb{Z}_m}$. Let $\mathcal{P}_x$ be an admissible monotone class of 4-graphs with $\text{dens}_2(\mathcal{P}_x) = x$. The graph H determines the complete $A^2 B_i B_j$ -profiles, while γ_i controls the link graphs indexed by $A \times B_i$.

Construction 4.2. Use a partition $V = A \dot{\cup} B_0 \dot{\cup} \dots \dot{\cup} B_{m-1}$. The following edge profiles are allowed.

- (S1) The induced 4-graph on A belongs to $\mathcal{P}_x$.
- (S2) For every edge $ij \in H$, all 4-sets of profile $A^2 B_i B_j$ are allowed.
- (S3) For each i , $a \in A$, and $y \in B_i$, the graph on B_{i-1} whose edges uv correspond to $\{a, u, v, y\} \in G$ is K_{s_i+1} -free. For $s_i = \infty$, no restriction is imposed.
- (S4) Every edge meeting four distinct parts is allowed.

(S5) *No other profile is allowed.*

Thus the resulting class is monotone; it is admissible because one may take the empty 4-graph on A and all link graphs empty.

The cyclic order and the auxiliary graph H play different roles, as summarized in Figure 1.

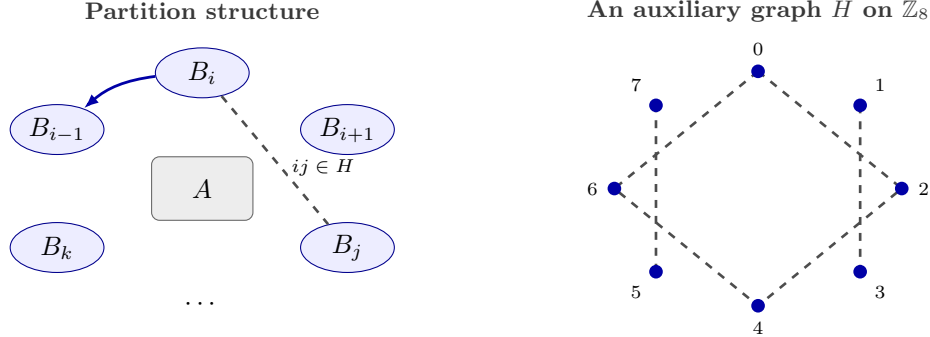


Figure 1: The satellite construction in Construction 4.2. The satellite subscripts are read cyclically. The directed arrow $B_i \rightarrow B_{i-1}$ denotes the K_{s_i+1} -free link on B_{i-1} indexed by $A \times B_i$, while dashed lines represent edges of H . On the right, the vertices of an example auxiliary graph on $\mathbb{Z}_8$ are placed in cyclic index order and drawn as solid circles; this placement does not require edges of H to join consecutive indices.

For the density, let $a, b_0, \dots, b_{m-1}$ be the asymptotic part weights, so that $a + \sum_i b_i = 1$. The *satellite support* of this weight vector is $\{i \in \mathbb{Z}_m : b_i > 0\}$. For pairs contained in a single part, the relevant normalized pair-degree expressions are

$$Q_A := xa^2 + 2 \sum_{ij \in H} b_i b_j, \quad \text{and} \quad Q_i := 2a\gamma_i b_i.$$

Here Q_i corresponds to pairs in B_{i-1} ; the associated link graphs are indexed by $A \times B_i$.

Define the satellite density function by

$$\Psi_{H,\gamma}(x) := \max \left\{ x, \max_{\substack{a, b_i \geq 0 \\ a + \sum b_i = 1}} \min\{Q_A, Q_0, \dots, Q_{m-1}\} \right\}.$$

We next record the endpoint estimate and the estimate at $x = 0$. The parameters below specify the target endpoint T , a value x_* for which we will prove $\Psi_{H,\gamma}(x_*) = T$, and an upper bound B for $\Psi_{H,\gamma}(0)$. Put $\alpha_i := \gamma_i^{-1}$ and

$$S := \sum_i \alpha_i, \quad K := \sum_{ij \in H} \alpha_i \alpha_j, \quad L := \max_i \sum_{j \in N_H(i)} \alpha_j, \\ \rho := 1 - \frac{K}{S}, \quad T := \frac{1}{2S}, \quad \kappa(\rho) := \frac{4(1-\rho)}{(2-\rho)^2}, \quad B := T\kappa(\rho), \quad \text{and} \quad x_* := 4\rho T.$$

For the estimate at $x = 0$, define the auxiliary quadratic bound, for $0 \leq u \leq 1$, by

$$\mathcal{E}(u) := 2\frac{K}{S}u^2 + 2Lu(1-u) + \frac{S}{2}(1-u)^2.$$

The following lemma is the interval-extension criterion used for every subsequent parameter choice.

Lemma 4.3. *Suppose that $0 \leq \rho \leq 1/4$, that $2u - \mathcal{E}(u) \geq 0$ and $\kappa(\rho)(\mathcal{E}(u) + 2u)^2 - 8u^2\mathcal{E}(u) \geq 0$ for every $u \in [\kappa(\rho), 1]$, and that $[0, M] \subseteq \Pi_{2,\infty}^4$, $B \leq M$, and $x_* \leq M < T$. Then $[0, T] \subseteq \Pi_{2,\infty}^4$.*

Proof. We first establish the density formula used below.

Claim 4.4. *The class resulting from Construction 4.2 has density $\Psi_{H,\gamma}(x)$. Moreover, $\Psi_{H,\gamma}$ is continuous and nondecreasing.*

Proof of the claim. For the upper bound, pass to a subsequence along which all part ratios of the witnessing partitions converge. If $a = 0$, choose i with $b_{i-1} > 0$. Every edge containing a pair from B_{i-1} then has a further vertex in A , so such a pair has degree $o(n^2)$. Suppose next that $a > 0$. If every $b_i > 0$, choose a pair in A whose degree within the induced 4-graph on A is at most $(x + o(1))\binom{|A|-2}{2}$. Its total normalized degree is at most $Q_A + o(1)$ because every additional edge containing two vertices of A lies in one of the complete profiles $A^2B_iB_j$. For fixed i , double count the edges of the link graphs on B_{i-1} indexed by $A \times B_i$. Turán's theorem bounds their average edge density by $\gamma_i + o(1)$, so some pair in B_{i-1} has normalized degree at most $Q_i + o(1)$.

If the satellite support is a nonempty proper subset of $\mathbb{Z}_m$, choose i with $b_{i-1} > 0$ and $b_i = 0$. Every completion of a pair from B_{i-1} then has its remaining two vertices in $A \times B_i$, and hence has degree $o(n^2)$. If all $b_i = 0$, choose a pair in the part A of linear size whose internal degree is at most $(x + o(1))\binom{|A|-2}{2}$; all of its external completions contribute only $o(n^2)$, so the value is at most x . These cases prove the upper bound in the max-min formula.

For the lower bound, the value x is realized asymptotically by taking $A = V$. If all of $Q_A, Q_0, \dots, Q_{m-1}$ are positive, then necessarily $a > 0$ and every $b_i > 0$. Place an asymptotically extremal member of $\mathcal{P}_x$ inside A . In every link graph use an independent uniformly random relabeling of a balanced complete s_i -partite graph when $s_i < \infty$, and use the complete graph when $s_i = \infty$. Chernoff concentration and a union bound over all pairs ensure simultaneously that every pair in B_{i-1} has normalized degree $Q_i - o(1)$ and every pair in A has normalized degree $Q_A - o(1)$.

It remains to check pairs meeting two distinct parts. Put $w := \min_i \gamma_i b_i$. For a pair meeting A and B_i , choose any neighbor j of i in H . The 4-sets of profile $A^2B_iB_j$ give normalized degree at least $2ab_j \geq 2aw$. For a pair meeting B_i and B_j , since $m \geq 3$, some satellite index t is distinct from i, j . The profiles meeting four distinct parts give normalized degree at least $2a \sum_{t \notin \{i,j\}} b_t \geq 2aw$. Here we used $b_t \geq \gamma_t b_t \geq w$ for every satellite index t . Thus every pair meeting two distinct parts has normalized degree at least $\min_i Q_i$. This proves the lower bound. The function $\Psi_{H,\gamma}$ is nondecreasing in x , and continuity follows from maximization of a continuous function over a compact simplex. $\square$

We now show that $\Psi_{H,\gamma}(x_*) = T$. At $a = 1/2$ and $b_i = T\alpha_i$, every satellite pair-degree expression equals $2 \cdot \frac{1}{2} \gamma_i T \alpha_i = T$ because $\gamma_i \alpha_i = 1$. Moreover, $x_*/4 + 2T^2K = \rho T + K/(2S^2) = \rho T + (1 - \rho)T = T$. For arbitrary weights, $\min_i \gamma_i b_i \leq (1 - a)/S$, so the minimum of the satellite pair-degree expressions is at most $2a(1 - a)/S \leq T$. Since $x_* = 4\rho T \leq T$, the first term in the definition of $\Psi_{H,\gamma}(x_*)$ is also at most T . Thus $\Psi_{H,\gamma}(x_*) = T$.

We next bound $\Psi_{H,\gamma}(0)$. For $x = 0$ and $a < 1$, write $b_i = (1 - a)y_i$, let $w := \min_i \gamma_i y_i$, and put $u := Sw$. The cases $a = 1$ or $w = 0$ give value zero. Otherwise $0 < u \leq 1$, and we write $y_i = w\alpha_i + z_i$, where $z_i \geq 0$ and $\sum_i z_i = 1 - u$. Expanding the weighted edge sum over H gives $2 \sum_{ij \in H} y_i y_j \leq \mathcal{E}(u)/S$. Indeed, the constant term is $2Ku^2/S^2$, the cross term is at most $2Lu(1 - u)/S$, and the Motzkin–Straus theorem [23] applied to the triangle-free graph H gives $2 \sum_{ij \in H} z_i z_j \leq (1 - u)^2/2$. After division by $T = 1/(2S)$, the expressions for pairs in a satellite part and for pairs in the central part are bounded by $4a(1 - a)u$ and $2(1 - a)^2\mathcal{E}(u)$, respectively. If $u < \kappa(\rho)$, the first expression is at most $u < \kappa(\rho)$. If $u \geq \kappa(\rho)$, then the first assumed inequality gives $\mathcal{E}(u) \leq 2u$, so the two expressions meet at a value of a at most $1/2$ and their minimum is maximized there. This maximum is $8u^2\mathcal{E}(u)/(2u + \mathcal{E}(u))^2$, which is at most $\kappa(\rho)$ by the second assumed inequality. Multiplication by T gives $\Psi_{H,\gamma}(0) \leq B$.

Every $x \in [0, x_*]$ belongs to $\Pi_{2,\infty}^4$ by hypothesis, so we may apply the satellite construction to a corresponding class. By Claim 4.4, the resulting class has density $\Psi_{H,\gamma}(x)$, and this function is continuous. Since $\Psi_{H,\gamma}(0) \leq B \leq M$ and $\Psi_{H,\gamma}(x_*) = T$, every value in $[B, T]$ belongs to $\Pi_{2,\infty}^4$. Taking the union with $[0, M]$ gives $[0, T]$. $\square$

To obtain an initial interval despite the discreteness of ordinary graph Turán densities, we use ordinary 3-graph classes as classes of vertex links.

Proposition 4.5. *For every integer $h \geq 4$ and every admissible monotone class $\mathcal{Q}$ of 3-graphs with $\text{dens}(\mathcal{Q}) = \beta$, there is an admissible monotone class of 4-graphs with minimum 2-degree density $2\beta/h^2$. Consequently, there is a constant $\eta_0 > 0$ such that $[0, \eta_0] \subseteq \Pi_{2,\infty}^4$.*

Proof. Define a class of 4-graphs by requiring its members G to admit a cyclic partition $V(G) = V_0 \dot{\cup} \dots \dot{\cup} V_{h-1}$ with the following properties.

(L1) For every i and every $y \in V_{i+1}$, the 3-graph $L_{i,y} := \{E \in \binom{V_i}{3} : E \cup \{y\} \in G\}$ belongs to $\mathcal{Q}$.

(L2) Every edge is either of the form $E \cup \{y\}$ with $E \in L_{i,y}$ for some i , or meets four distinct parts.

The source part of an edge of type (L1) is unique, so this class is monotone. It is admissible because every empty link belongs to $\mathcal{Q}$.

For the lower bound, split the vertices equally among the h parts. For each i and $y \in V_{i+1}$, choose an independent uniformly random relabeling of a member $H \in \mathcal{Q}$ with edge density $\beta - o(1)$. For a fixed pair $S \subseteq V_i$, its expected pair-degree inside one such vertex link is $3|H|/\binom{|V_i|}{2} = (\beta - o(1))(|V_i| - 2)$. Summing over $y \in V_{i+1}$, applying Hoeffding's inequality, and taking a union bound over all $O(n^2)$ pairs contained in one part gives $d_G(S) \geq (\beta - o(1))n^2/h^2$. Since $\binom{n-2}{2} = (1/2 + o(1))n^2$, the normalized degree is at least $2\beta/h^2 - o(1)$.

If a pair meets two distinct parts, then choosing one vertex from each of two further distinct parts produces an edge meeting four distinct parts. Its normalized degree is at least $(h-2)_2/h^2 - o(1) \geq 2/h^2 - o(1) \geq 2\beta/h^2 - o(1)$.

For the upper bound, consider any sequence of witnessing partitions with part sizes $w_0, \dots, w_{h-1}$. After taking a subsequence, assume that $w_i/n \rightarrow z_i$ for every i . For each i , double counting the incidences between pairs in V_i and their completions in $V_i \times V_{i+1}$ gives

$$\sum_{S \in \binom{V_i}{2}} d_G(S) = \sum_{y \in V_{i+1}} 3|L_{i,y}|.$$

If the support of (z_i) is proper, choose i with $z_i > 0$ and $z_{i+1} = 0$. The trivial bound $|L_{i,y}| \leq \binom{w_i}{3}$ and the displayed identity give a pair of degree at most $(w_i - 2)w_{i+1} = o(n^2)$. If the support is full, all w_i tend to infinity. For fixed $\varepsilon > 0$, the definition of $\text{dens}(\mathcal{Q}) = \beta$ gives $|L_{i,y}| \leq (\beta + \varepsilon)\binom{w_i}{3}$ for all sufficiently large n . The same identity therefore gives, for every i , a pair in V_i of degree at most $(\beta + \varepsilon)(w_i - 2)w_{i+1}$. Hence

$$\limsup_{n \rightarrow \infty} \frac{\delta_2(G)}{n^2} \leq (\beta + \varepsilon) \min_i z_i z_{i+1} \leq (\beta + \varepsilon) \left(\prod_i z_i \right)^{2/h} \leq \frac{\beta + \varepsilon}{h^2}.$$

Dividing by $\binom{n-2}{2}$ and letting $\varepsilon \rightarrow 0$ proves the formula.

It remains to prove the interval assertion. By [Theorem 2.1](#), every interval $[2c_3/h^2, 2/h^2]$ lies in $\Pi_{2,\infty}^4$. For all sufficiently large h , consecutive intervals overlap because $2/(h+1)^2 \geq 2c_3/h^2$. Their union, together with the density zero, therefore contains $[0, \eta_0]$ for some $\eta_0 > 0$. $\square$

4.2 Extending the interval to 3/38

[Proposition 4.5](#) supplies an interval adjacent to zero. We now iterate the satellite extension criterion, first with homogeneous parameters and then with heterogeneous ones, to reach 3/38.

Proposition 4.6. $[0, 3/38] \subseteq \Pi_{2,\infty}^4$.

The satellite extension criterion reduces the proof to two tasks. First, we prove the two inequalities in [Lemma 4.3](#) uniformly over the parameter ranges used in the constructions below. Second, we provide finitely many rational *certificate rows*. Each row specifies m , the auxiliary graph H either explicitly or through an allowed family, the satellite parameters $s_0, \dots, s_{m-1}$, and an endpoint M for which $[0, M]$ has already been realized. The quantities S, K, L, ρ, T, B , and x_* are then determined by these data. The inequalities $B \leq M$ and $x_* \leq M < T$ certify one application of [Lemma 4.3](#), extending $[0, M]$ to $[0, T]$. The first task is analytic, while the certificate rows are checked by exact rational arithmetic.

We first treat homogeneous parameters. Suppose all satellite parameters are equal to $\gamma := 1 - 1/s$, where $s \in \{2, 3, \dots\} \cup \{\infty\}$ and $1/\infty := 0$, and H is a union of nontrivial paths on m vertices. Multiplying the general quadratic bound by γ and using $\Delta(H) \leq 2$ gives

$$\mathcal{E}_{\text{hom}}(u) := 2\gamma(1 - \rho)u^2 + 4u(1 - u) + \frac{m}{2}(1 - u)^2.$$

The two scaled auxiliary polynomials are

$$D_{\text{hom}}(u) := 2\gamma u - \mathcal{E}_{\text{hom}}(u), \quad \text{and} \quad G_{\text{hom}}(u) := \kappa(\rho)(\mathcal{E}_{\text{hom}}(u) + 2\gamma u)^2 - 8\gamma u^2 \mathcal{E}_{\text{hom}}(u).$$

The degree bound $\Delta(H) \leq 2$ gives $0 \leq \gamma \mathcal{E}(u) \leq \mathcal{E}_{\text{hom}}(u)$. If $D_{\text{hom}}(u) \geq 0$, then $\mathcal{E}_{\text{hom}}(u)/\gamma \leq 2u$. For fixed $u \geq \kappa(\rho)$ and $0 \leq \xi \leq 2u$, the derivative with respect to ξ satisfies

$$\frac{\partial}{\partial \xi} (\kappa(\rho)(\xi + 2u)^2 - 8u^2 \xi) = 2\kappa(\rho)(\xi + 2u) - 8u^2 \leq 8u(\kappa(\rho) - u) \leq 0.$$

Moreover, $G_{\text{hom}}(u)$ is γ^2 times this expression evaluated at $\xi = \mathcal{E}_{\text{hom}}(u)/\gamma$. Since $\mathcal{E}(u) \leq \mathcal{E}_{\text{hom}}(u)/\gamma$, its value at $\xi = \mathcal{E}(u)$ is no smaller. Thus the two scaled inequalities below imply the two inequalities in [Lemma 4.3](#) for $\mathcal{E}$.

Lemma 4.7. *If $1/7 \leq \rho \leq 1/4$, $4/5 \leq \gamma \leq 1$, and $8 \leq m \leq 55$, then $D_{\text{hom}}(u) \geq 0$ and $G_{\text{hom}}(u) \geq 0$ for every $u \in [\kappa(\rho), 1]$.*

Proof. Set $u := \kappa(\rho) + (1 - \kappa(\rho))z$, where $0 \leq z \leq 1$. The auxiliary numerators and the degree-two Bernstein coefficients of D_{hom} are

$$\begin{aligned} A_D &:= \gamma(-48\rho^2 + 112\rho - 64) + m\rho^3 + 32\rho(1 - \rho), & \text{and} & \quad C_D := 7\gamma\rho - 8\gamma + 2\rho; \\ d_0 &= -\frac{\rho A_D}{2(2 - \rho)^4}, & d_1 &= -\frac{\rho C_D}{(2 - \rho)^2}, & \text{and} & \quad d_2 = 2\gamma\rho. \end{aligned}$$

The parameter bounds give

$$A_D \leq -\frac{156}{5} + \frac{55}{64} + 6 = -\frac{7789}{320} < 0, \quad \text{and} \quad C_D \leq -\frac{9}{2} < 0.$$

Hence all three coefficients are nonnegative.

The degree-four Bernstein coefficients of G_{hom} are

$$g_0 = \frac{\rho^2(1 - \rho)A_D^2}{(2 - \rho)^{10}}, \quad g_1 = \frac{2\rho^2(1 - \rho)Q_1}{(2 - \rho)^8}, \quad g_2 = \frac{2\rho^2Q_2}{3(2 - \rho)^6}, \quad g_3 = \frac{8\gamma\rho^2(\gamma - (\gamma + 1)\rho)}{(2 - \rho)^3}, \quad \text{and} \quad g_4 = 0,$$

where

$$\begin{aligned} Q_1 &= \gamma^2(384 - 960\rho + 784\rho^2 - 208\rho^3) - \gamma m\rho^3(8 - 5\rho) \\ &\quad + \gamma(-256\rho^3 + 640\rho^2 - 384\rho) + 2m\rho^4 + 64\rho^2(1 - \rho), \\ Q_2 &= \gamma^2(192 - 480\rho + 388\rho^2 - 100\rho^3) - \gamma m\rho^3(2 - \rho) \\ &\quad + \gamma(-112\rho^3 + 304\rho^2 - 192\rho) + 16\rho^2(1 - \rho). \end{aligned}$$

On $0 \leq \rho \leq 1/4$, the two positive γ^2 -factors and the two remaining γ -factors are decreasing, whereas $\rho^3(8 - 5\rho)$ and $\rho^3(2 - \rho)$ are increasing. These monotonicities follow by differentiating the six cubic polynomials, and all the relevant extrema are therefore attained at $\rho = 1/4$. The two positive factors are at least $759/4$ and $1515/16$, the two factors multiplying the negative m -terms are at most $27/256$ and $7/256$, and the remaining factors are at least -60 and $-123/4$, respectively. Consequently,

$$Q_1 \geq \frac{16}{25} \frac{759}{4} - 55 \frac{27}{256} - 60 = \frac{356091}{6400} > 0 \quad \text{and} \quad Q_2 \geq \frac{16}{25} \frac{1515}{16} - 55 \frac{7}{256} - \frac{123}{4} = \frac{36283}{1280} > 0.$$

Finally, $\gamma - (\gamma + 1)\rho \geq 4/5 - 9/20 = 7/20 > 0$. All Bernstein coefficients are nonnegative, so both polynomials are nonnegative on the stated interval. $\square$

We next treat heterogeneous parameters. For general satellite parameters, the identity $K/S = 1 - \rho$ rewrites the quadratic bound as $\mathcal{E}(u) = 2(1 - \rho)u^2 + 2Lu(1 - u) + S(1 - u)^2/2$.

Lemma 4.8. *Suppose that $0 < \rho \leq 1/4$. The two inequalities in [Lemma 4.3](#) hold on $[\kappa(\rho), 1]$ if either $S \leq 10$ and $L \leq 5/2$, or $S \leq 31/5$ and $L \leq 3$.*

Proof. After applying the substitution $u = \kappa(\rho) + (1 - \kappa(\rho))z$ to this expression, the degree-two Bernstein coefficients and their auxiliary numerators are

$$d_0 = -\frac{\rho A_0}{2(2 - \rho)^4}, \quad d_1 = -\frac{\rho C_0}{(2 - \rho)^2}, \quad d_2 = 2\rho,$$

$$A_0 := -16L\rho^2 + 16L\rho + S\rho^3 - 48\rho^2 + 112\rho - 64, \quad \text{and} \quad C_0 := L\rho + 7\rho - 8.$$

For the first parameter range, the bounds give

$$A_0 \leq -39 + \frac{15}{2} + \frac{5}{32} = -\frac{1003}{32} < 0, \quad \text{and} \quad C_0 \leq -\frac{45}{8} < 0.$$

The degree-four coefficients are

$$g_0 = \frac{\rho^2(1 - \rho)A_0^2}{(2 - \rho)^{10}}, \quad g_1 = \frac{2\rho^2(1 - \rho)R_1}{(2 - \rho)^8}, \quad g_2 = \frac{2\rho^2R_2}{3(2 - \rho)^6}, \quad g_3 = \frac{4\rho^2(2 - (L + 2)\rho)}{(2 - \rho)^3}, \quad \text{and} \quad g_4 = 0,$$

where

$$R_1 = -16L^2\rho^3 + 16L^2\rho^2 + LS\rho^4 - 128L\rho^3 + 320L\rho^2 - 192L\rho$$

$$+ 5S\rho^4 - 8S\rho^3 - 208\rho^3 + 784\rho^2 - 960\rho + 384,$$

$$R_2 = -4L^2\rho^3 + 4L^2\rho^2 - 56L\rho^3 + 152L\rho^2 - 96L\rho$$

$$+ S\rho^4 - 2S\rho^3 - 100\rho^3 + 388\rho^2 - 480\rho + 192.$$

Group the terms by their dependence on L and S . The L^2 - and LS -groups are nonnegative. Each remaining polynomial in ρ is decreasing on $[0, 1/4]$, so its minimum is attained at $\rho = 1/4$; for the negative L - and S -groups, the minimum over the first parameter range also occurs at $L = 5/2$ or $S = 10$, respectively. Substitution at these endpoints bounds the constant, remaining L -, and remaining S -groups below by $759/4, -75, -135/128$ for R_1 , and by $1515/16, -615/16, -35/128$ for R_2 . Therefore

$$R_1 \geq \frac{759}{4} - 75 - \frac{135}{128} = \frac{14553}{128} > 0, \quad \text{and} \quad R_2 \geq \frac{1515}{16} - \frac{615}{16} - \frac{35}{128} = \frac{7165}{128} > 0.$$

Also, $2 - (L + 2)\rho \geq 2 - (9/2)(1/4) = 7/8 > 0$. The endpoint coefficients g_0 and g_4 are nonnegative as well, so every Bernstein coefficient is nonnegative for the first parameter range.

For the second parameter range, the two nontrivial degree-two factors satisfy

$$A_0 \leq 9 + \frac{31}{320} + 25 - 64 = -\frac{9569}{320} < 0, \quad \text{and} \quad C_0 \leq -\frac{11}{2} < 0.$$

For the quartic factors, direct differentiation gives

$$\partial_S R_1 = \rho^3((L + 5)\rho - 8) < 0 \quad \text{and} \quad \partial_S R_2 = \rho^3(\rho - 2) < 0,$$

$$\rho^{-1}\partial_L R_1 \leq -\frac{28129}{320} < 0 \quad \text{and} \quad \rho^{-1}\partial_L R_2 \leq -52 < 0.$$

It remains to vary ρ after setting $L = 3$ and $S = 31/5$. The resulting polynomials are

$$R_1 = 384 - 1536\rho + 1888\rho^2 - \frac{3928}{5}\rho^3 + \frac{248}{5}\rho^4,$$

$$R_2 = 192 - 768\rho + 880\rho^2 - \frac{1582}{5}\rho^3 + \frac{31}{5}\rho^4.$$

Their second derivatives are positive on $[0, 1/4]$, while $R_1'(1/4) = -3681/5$ and $R_2'(1/4) = -6191/16$. Hence both are decreasing there, so over the second parameter range both attain their minimum at $(L, S, \rho) = (3, 31/5, 1/4)$. At this point, the values are $R_1 = 16947/160 > 0$ and $R_2 = 64103/1280 > 0$. The final quartic factor satisfies $2 - (L + 2)\rho \geq 3/4 > 0$, so every Bernstein coefficient is nonnegative throughout this range as well. $\square$

Proof of Proposition 4.6. We first show how an arbitrary interval adjacent to zero can be extended to $[0, 1/114]$. In the homogeneous specialization, let $m \geq 3$ and $c \geq 1$ be integers with $2c \leq m$, and let H be

a union of c nontrivial paths on m vertices. Then $|H| = m - c$. For $s \in \{2, 3, \dots\} \cup \{\infty\}$, put $\gamma := 1 - 1/s$, $\rho := (\gamma m - |H|)/(\gamma m)$, and $T := \gamma/(2m)$. For every integer $h \geq 57$, put $q_h := \lceil 2\sqrt{h+1} - 2 \rceil$, specialize to $m = h$, $c = q_h$, and $\gamma = 1$, and let H be a union of q_h nontrivial paths on h satellites.

The parameter range here is not covered by [Lemma 4.7](#), because $m = h \geq 57$. We therefore verify its Bernstein coefficients directly. Here $\rho = q_h/h$, and the definition of q_h gives $q_h < 2\sqrt{h+1} - 1$ and $q_h^2 < 4h$. Moreover, $2\sqrt{h+1} - 1 \leq h/4$ for $h \geq 57$; after squaring, this is equivalent to $h^2 - 56h - 48 \geq 0$. Hence $0 < \rho \leq 1/4$ and $h\rho^2 < 4$. With $\gamma = 1$ and $m = h$, the coefficient numerators from the proof of [Lemma 4.7](#) satisfy

$$\begin{aligned} A_D &= -64 + 144\rho - 80\rho^2 + h\rho^3 \leq -27 \quad \text{and} \quad C_D = 9\rho - 8 \leq -\frac{23}{4}, \\ Q_1 &= 384 - 1344\rho + 1488\rho^2 - 528\rho^3 - h\rho^3(8 - 7\rho) \geq \frac{127}{4}, \\ Q_2 &= 192 - 672\rho + 708\rho^2 - 228\rho^3 - h\rho^3(2 - \rho) \geq \frac{295}{16}. \end{aligned}$$

Here we used $h\rho^3 < 1$; for the last two bounds we dropped their positive quadratic terms and used $h\rho^3(8 - 7\rho) < 8$ and $h\rho^3(2 - \rho) < 2$. Finally, $1 - 2\rho \geq 1/2$. Thus every Bernstein coefficient of D_{hom} and G_{hom} is nonnegative, which proves the two inequalities in [Lemma 4.3](#) for these parameter choices. The inequality $q_h \leq h/4$ also ensures that a union of q_h nontrivial paths on h vertices exists.

It remains to verify the overlap conditions needed for the implication $[0, 1/(2(h+1))] \subseteq \Pi_{2,\infty}^4 \Rightarrow [0, 1/(2h)] \subseteq \Pi_{2,\infty}^4$. They reduce to $q_h^2 + 4q_h \geq 4h$ and $4q_h(h+1) \leq h^2$. The inequality $q_h^2 + 4q_h \geq 4h$ follows directly from the definition of q_h . For $57 \leq h \leq 63$, the inequality $4q_h(h+1) \leq h^2$ is verified by an exact seven-case check, with minimum slack $h^2 - 4q_h(h+1) = 1$ at $h = 57$. Beyond this range, separate the cases $q_h = 2u + 1$ and $q_h = 2u$. The ceiling definition gives $h \geq (u+1)^2$ in the odd case and $h \geq u(u+1)$ in the even case. For fixed q_h , the slack $h^2 - 4q_h(h+1)$ is increasing throughout the relevant range, so it is enough to substitute these lower bounds for h . The resulting slacks are

$$u^4 - 4u^3 - 14u^2 - 20u - 7 > 0 \quad \text{for } u \geq 7 \quad \text{and} \quad u(u^3 - 6u^2 - 7u - 8) > 0 \quad \text{for } u \geq 8,$$

respectively. They are positive at $u = 7$ and $u = 8$, and their forward differences are positive thereafter.

Choose h_0 so large that $1/(2(h_0+1)) \leq \eta_0$, where η_0 is supplied by [Proposition 4.5](#). Iterating downward in h finitely many times gives $[0, 1/114] \subseteq \Pi_{2,\infty}^4$.

We next complete the homogeneous stage with the 113 certificate rows recorded in `weighted_path_chain_certificate.txt`: four exceptional rows reach $1/106$, and the remaining 109 reach $1/20$. The proof of [Lemma 4.7](#) gives the common Bernstein-polynomial argument, while the certificate rows verify only $B \leq M$ and $x_* \leq M < T$. These rows are checked by `verify_pi2_4_exact.py`.

For the heterogeneous stage, the 57 rows in `heterogeneous_satellite_chain_certificate.txt` are used in three finite batches connected by two infinite parametric families. These rows are checked by `verify_pi2_4_exact.py`. Put $T_t^{(8)} := (t-1)/(16t)$ and $T_t^{(7)} := (t-1)/(14t)$; these sequences increase to $1/16$ and $1/14$, respectively. They occur in the order

$$\frac{1}{20} \xrightarrow{\text{rows 1-17}} \frac{13525214010}{236953260851} > T_{11}^{(8)} \nearrow \frac{1}{16} \xrightarrow{\text{rows 18-36}} \frac{999}{14000} > T_{20}^{(7)} \nearrow \frac{1}{14} \xrightarrow{\text{rows 37-57}} \frac{3996}{50315} > \frac{3}{38}.$$

The first parametric family uses $H = P_6 \dot{\cup} P_2$ on eight satellites with homogeneous parameter $s = t$, and the second uses $H = P_7$ on seven satellites. Both satisfy the hypotheses of [Lemma 4.8](#). Indeed, their respective triples (S, L, ρ) are

$$\left(\frac{8t}{t-1}, \frac{2t}{t-1}, \frac{t-4}{4(t-1)} \right) \quad \text{and} \quad \left(\frac{7t}{t-1}, \frac{2t}{t-1}, \frac{t-7}{7(t-1)} \right).$$

Writing $B_t^{(m)}$ and $x_{*,t}^{(m)}$ for the values of B and x_* in the m -satellite parameter choice, we verify the two overlap inequalities required by [Lemma 4.3](#) separately for the two families. For the first family, every integer $t \geq 12$ satisfies

$$T_{t-1}^{(8)} - B_t^{(8)} = \frac{(t+2)(t^2 - 12t + 8)}{16(t-1)(7t-4)^2} \geq 0, \quad \text{and} \quad T_{t-1}^{(8)} - x_{*,t}^{(8)} = \frac{3t-4}{16t(t-1)} > 0.$$

For the second family, every integer $t \geq 21$ satisfies

$$T_{t-1}^{(7)} - B_t^{(7)} = \frac{(t+5)(t^2 - 21t + 14)}{14(t-1)(13t-7)^2} \geq 0, \quad \text{and} \quad T_{t-1}^{(7)} - x_{*,t}^{(7)} = \frac{3t^2 + 18t - 28}{98t(t-1)} > 0.$$

The transversal-link construction realizes the limiting endpoint densities, namely $1/16 = (1/2)(2)_2/4^2$ and $1/14 = (27/28)(3)_2/9^2$.

For the final heterogeneous row, take the following labeled graph and satellite parameters:

$$H := \{\{1, 4\}, \{0, 2\}, \{2, 3\}, \{3, 5\}, \{0, 5\}\}, \quad \text{and} \quad (s_0, \dots, s_5) = (1000, 9, 1000, 1000, 7, 1000),$$

which is a labeled copy of $C_4 \dot{\cup} P_2$. For this choice, $T = 3996/50315$ and $\rho = 15571349/100529370$, while

$$B = \frac{2713222077312672}{34405572219986881} \leq \frac{999}{12662} \quad \text{and} \quad x_* = \frac{124570792}{2531599225} \leq \frac{999}{12662}.$$

Finally, $999/12662 < 3/38 < 3996/50315$; the two positive differences are $6/120289$ and $903/1911970$, respectively. Thus the interval obtained contains $[0, 3/38]$. $\square$

4.3 The D_{25} extension to $2/25$

[Proposition 4.6](#) leaves a narrow gap before the high-density transfer stage. We close it with a 25-satellite construction whose continuous density map connects $3/38$ to $2/25$.

Proposition 4.9. $[0, 2/25] \subseteq \Pi_{2,\infty}^4$.

Proof. We begin with the template and its density map.

On $\mathbb{Z}_{25}$, define the graph H by

$$H := \{\{i, i+d\} : i \in \mathbb{Z}_{25} \text{ and } d \in \{3, 8, 10\}\} \cup \{\{0, 1\}\}.$$

The first set is interpreted as an undirected Cayley graph, so it has 75 edges and $|H| = 76$. The graph is triangle-free. Indeed, the difference set $\mathcal{D} := \{\pm 3, \pm 8, \pm 10\}$ satisfies $\mathcal{D} \cap (\mathcal{D} + \mathcal{D}) = \emptyset$ in $\mathbb{Z}_{25}$, and a triangle using the extra edge 01 would require a common neighbor of 0 and 1 , excluded by $\mathcal{D} \cap (1 + \mathcal{D}) = \emptyset$. We record this because H controls the degree expression for pairs in the central part, just as the auxiliary triangle-free graphs do in the satellite construction; the sharper quadratic estimate below will use its adjacency matrix directly.

Let $\mathcal{P}_x$ be an admissible monotone class of 4-graphs with density x . Use a partition $V = A \dot{\cup} B_0 \dot{\cup} \dots \dot{\cup} B_{24}$, where A is the central part and $B_0, \dots, B_{24}$ are the satellite parts, and allow the following profiles.

- (D1) $G[A] \in \mathcal{P}_x$;
- (D2) all profiles $A^2 B_i B_j$ with $ij \in H$;
- (D3) all profiles $AB_i^2 B_{i+t}$ with $t \in \{1, 2, 3, 4\}$;
- (D4) all profiles meeting four distinct parts;
- (D5) no other profiles.

The graph H and the preceding profile list form the finite template. Since $\mathcal{P}_x$ is admissible and monotone, the resulting class is also admissible and monotone. We call this the D_{25} construction applied to $\mathcal{P}_x$.

Let a and b_i denote the asymptotic weights of A and B_i , respectively. Thus $a + \sum_i b_i = 1$. For pairs contained in a single part, the relevant normalized pair-degree expressions are

$$P_A := xa^2 + 2 \sum_{ij \in H} b_i b_j, \quad \text{and} \quad P_i := 2a(b_{i+1} + b_{i+2} + b_{i+3} + b_{i+4}).$$

The upper estimates below use only these expressions. To prove continuity, however, we must also account for changes of support and for every pair type meeting two distinct parts.

Define $\Psi_{25}(x)$ by the following finite max–min problem. For each nonempty set J of parts, consider the closed simplex of weight vectors whose coordinates outside J vanish. Each unordered pair type determined by parts in J has a continuous normalized pair-degree polynomial in x and the part weights, obtained by including all allowed edges of every external profile and normalizing by $\binom{n-2}{2}$; for the type AA , the internal contribution is xz_A^2 . On each simplex, maximize the minimum of these pair-degree polynomials, and then take the maximum over all choices of J .

The class resulting from the D_{25} construction has density $\Psi_{25}(x)$. Indeed, fix a convergent sequence of part weights and let K be the support of its limit. If $A \in K$, then A has linear size and the definition of $\text{dens}_2(\mathcal{P}_x)$ supplies a pair in A whose internal normalized degree is at most $x + o(1)$. For every other supported pair type, averaging gives a pair whose normalized degree is at most the corresponding normalized pair-degree polynomial. This proves the upper bound. Conversely, for a fixed limiting weight vector with support K , take an asymptotically extremal member of $\mathcal{P}_x$ inside A when $A \in K$ and include all allowed edges of every external profile. Every supported pair type then attains its normalized pair-degree polynomial up to $o(1)$, proving the lower bound.

If a weight vector in the simplex indexed by J has smaller support K , deleting its zero-weight parts removes pair-type constraints and cannot decrease the minimum. The same vector is considered in the simplex indexed by K , so allowing boundary points does not change the maximum. For each fixed J , the minimum of finitely many normalized pair-degree polynomials is continuous, and its maximum over the compact simplex is continuous in x . Since there are only finitely many choices of J , Ψ_{25} is continuous.

We first handle proper satellite supports, where at least one but not all satellite parts have positive weight.

For every nonempty $I \subseteq \mathbb{Z}_{25}$, define $\omega(I)$ to be the supremum, over all positive weights $(y_i)_{i \in I}$ summing to one, of

$$\min_{i \in I} (y_{i+1} + y_{i+2} + y_{i+3} + y_{i+4}),$$

where $y_j = 0$ outside I .

Claim 4.10. *If $\emptyset \neq I \subsetneq \mathbb{Z}_{25}$, then $\omega(I) \leq 3/19$.*

Proof. We first note support monotonicity. If $\emptyset \neq J \subseteq I$, then $\omega(J) \leq \omega(I)$. It is enough to prove this when $I = J \cup \{t\}$. If $\omega(J) = 0$, there is nothing to prove. Otherwise every cyclic gap between consecutive points of J is at most four. Let u, v be consecutive support points with t between them. Every support point contributing to the forward four-term sum at u also contributes to the forward four-term sum at t . Extend a nearly optimal weighting on J by assigning an arbitrarily small positive weight to t and renormalizing. The new four-term sum at t is no smaller, up to an arbitrarily small error, than the old sum at u , while all other old sums can only gain weight before renormalization. This proves monotonicity.

It therefore suffices to consider a support missing exactly one point. Rotate so that the missing point is 0, and average the forward four-term sums over the 19 nonzero indices not divisible by four. Every nonzero index occurs in exactly three of these sums, so their total is $3 \sum_{j=1}^{24} y_j = 3$. Their average is therefore $3/19$, and at least one of them is at most $3/19$. $\square$

Consequently, if the total satellite weight is $s := 1 - a$ and the normalized satellite weights have nonempty proper support, Claim 4.10 gives a pair contained in a satellite part whose normalized degree is at most $2as(3/19) \leq 3/38$. If the satellite support is empty, the value is at most x . Thus for $x \leq 3/38$, every configuration with nonempty proper satellite support has value at most $3/38$.

We next prove the left-endpoint estimate $\Psi_{25}(0) \leq 3/38$. Assume first that all satellites have positive weight, and write

$$b_i = sy_i, \quad W_i := y_{i+1} + y_{i+2} + y_{i+3} + y_{i+4}, \quad w := \min_i W_i, \quad \text{and} \quad \Phi_H(\mathbf{y}) := 2 \sum_{ij \in H} y_i y_j,$$

where $a + s = 1$ and $\sum_i y_i = 1$. At $x = 0$, we have $P_A = s^2 \Phi_H(\mathbf{y})$ and $\min_i P_i = 2asw$, so the degree expressions for pairs contained in a single part bound the normalized minimum pair degree by

$$\min\{s^2 \Phi_H(\mathbf{y}), 2asw\}. \tag{4.1}$$

If $w < 3/19$, then the second term is at most $w/2 < 3/38$. Since $\sum_i W_i = 4$, we have $w \leq 4/25$, so it remains to consider $3/19 \leq w \leq 4/25$. Put $\tau := 4/25 - w$, so $0 \leq \tau \leq 1/475$.

Claim 4.11. *In this range, $\Phi_H(\mathbf{y}) \leq 152/625 + (24/25)\tau + 297\tau^2$.*

Proof. Let $\mathbf{C}$ be the 25×25 cyclic four-sum matrix defined by $(\mathbf{C}\mathbf{y})_i = y_{i+1} + y_{i+2} + y_{i+3} + y_{i+4}$, and let $\mathbf{1}$ be the all-ones column vector. The matrix $\mathbf{C}$ is invertible, with $\det \mathbf{C} = 4$. If $\tau = 0$, then $\mathbf{C}\mathbf{y} = (4/25)\mathbf{1}$, and invertibility together with $\mathbf{C}\mathbf{1} = 4\mathbf{1}$ gives $\mathbf{y} = \mathbf{1}/25$; the claimed bound is then an equality. We may therefore assume that $\tau > 0$. Since $\sum_i W_i = 4$, the vector $\mathbf{p}$ defined by $p_i := (W_i - w)/(25\tau)$ is a probability vector. Set $\mathbf{J} := \mathbf{1}\mathbf{1}^\top$ and define $\mathbf{B} := 25\mathbf{C}^{-1} - \mathbf{J}/4$. A direct inversion gives

$$\mathbf{B}_{ij} = \begin{cases} -19, & j - i \in \{0, 4, 8, 12, 16, 20\}, \\ 6, & \text{otherwise} \end{cases}.$$

Since $\mathbf{C}\mathbf{y} = w\mathbf{1} + 25\tau\mathbf{p}$, $\mathbf{C}^{-1}\mathbf{1} = \mathbf{1}/4$, and $\mathbf{J}\mathbf{p} = \mathbf{1}$, we have

$$\mathbf{y} = \frac{1}{25}\mathbf{1} + \tau\mathbf{B}\mathbf{p}. \quad (4.2)$$

To bound the quadratic term in (4.3), let $\mathbf{A}_H$ be the adjacency matrix of H , put $\mathbf{R} := \mathbf{B}^\top \mathbf{A}_H \mathbf{B}$, and define

$$L_0 := \{0, 1, 4, 5, 8, 9, 12, 13, 16, 17, 20, 21\}, \quad \text{and} \quad \mu := \sum_{i \in L_0} p_i.$$

Expansion of (4.2) gives

$$\Phi_H(\mathbf{y}) = \frac{152}{625} + \tau \left(\frac{24}{25} - 2\mu \right) + \tau^2 \mathbf{p}^\top \mathbf{R} \mathbf{p}. \quad (4.3)$$

We use the following quadratic-form bound for probability vectors.

$$\mathbf{p}^\top \mathbf{R} \mathbf{p} \leq 297 + 950\mu. \quad (4.4)$$

Let χ be the indicator vector of L_0 and define

$$\mathbf{K}_0 := 297\mathbf{J} + 475(\chi\mathbf{1}^\top + \mathbf{1}\chi^\top) - \mathbf{R}.$$

All negative entries of $\mathbf{K}_0$ are exactly the adjacent pairs of the three disjoint paths $6 - 3 - 18 - 15$, $10 - 7 - 22 - 19$, and $2 - 24 - 14 - 11$. On a path written $a - b - c - d$, the corresponding principal block of $\mathbf{K}_0/625$ has quadratic form $(p_a - p_b + p_c - p_d)^2 + 6p_a p_d \geq 0$. Every entry joining distinct path blocks or a path block to the complement is nonnegative. Therefore $\mathbf{p}^\top \mathbf{K}_0 \mathbf{p} \geq 0$ for every nonnegative $\mathbf{p}$, which is exactly (4.4) because $\sum_i p_i = 1$.

Substitution into (4.3) yields

$$\Phi_H(\mathbf{y}) \leq \frac{152}{625} + \frac{24}{25}\tau + 297\tau^2 + \mu\tau(950\tau - 2).$$

The last term is nonpositive because $\tau \leq 1/475$. □

We now apply the claim to the full-support range. Put $u := 25\tau \in [0, 1/19]$, so $w = (4 - u)/25$, and define $\bar{\Phi}(u) := (297u^2 + 24u + 152)/625$. A direct factorization gives $2w - \bar{\Phi}(u) = (11u + 6)(8 - 27u)/625 > 0$. For fixed $\xi < 2w$, the maximum over $a + s = 1$ of (4.1) is attained when its two terms are equal and is $\phi_{25}(\xi, w) := 4\xi w^2/(\xi + 2w)^2$. This expression is increasing in ξ , so the claim gives $\phi_{25}(\Phi_H(\mathbf{y}), w) \leq \phi_{25}(\bar{\Phi}(u), w)$. The value at the upper bound and its difference from $19/242$ are

$$\begin{aligned} \phi_{25}(\bar{\Phi}(u), w) &= \frac{4(u - 4)^2(297u^2 + 24u + 152)}{(297u^2 - 26u + 352)^2}, \\ \phi_{25}(\bar{\Phi}(u), w) - \frac{19}{242} &= -\frac{25u(55539u^3 + 79332u^2 - 23028u + 18304)}{242(297u^2 - 26u + 352)^2}. \end{aligned}$$

The cubic factor is positive on $[0, 1/19]$, since $18304 - 23028u \geq 18304 - 23028/19 > 0$. This proves the sharper bound on the full-support range $w \geq 3/19$.

The case $w < 3/19$ was handled above. Nonempty proper satellite supports have value at most $3/38$ by Claim 4.10, and the empty satellite support has value zero at $x = 0$. These cases prove $\Psi_{25}(0) \leq 3/38$.

We finally prove the right-endpoint estimate. By [Proposition 4.6](#), the interval $[0, 48/625]$ is already realized because $48/625 < 3/38$. At $x = 48/625$, take the balanced weights $a = 1/2$ and $b_i = 1/50$. Every pair-degree expression P_i for a satellite part equals $2/25$, while the expression for pairs in the central part is $P_A = 48/(4 \cdot 625) + 2 \cdot 76(1/50)^2 = 12/625 + 38/625 = 2/25$. Every pair meeting two distinct parts has larger degree because of the profiles meeting four distinct parts. Indeed, the normalized contributions from these profiles to a pair meeting A and B_i and to a pair meeting two satellite parts are at least, respectively,

$$2 \binom{24}{2} \frac{1}{50^2} = \frac{552}{2500} > \frac{2}{25}, \quad \text{and} \quad 2 \binom{23}{2} \frac{1}{50^2} = \frac{506}{2500} > \frac{2}{25}.$$

This proves $\Psi_{25}(48/625) \geq 2/25$.

For the reverse inequality, suppose first that all satellites have positive weight. Averaging their forward four-term sums gives $\min_i (b_{i+1} + b_{i+2} + b_{i+3} + b_{i+4}) \leq 4(1-a)/25$, and hence $\min_i P_i \leq 2a(1-a)(4/25) \leq 2/25$. A nonempty proper satellite support has value at most $3/38 < 2/25$ by [Claim 4.10](#), whereas the empty satellite support has value at most $48/625 < 2/25$. Therefore $\Psi_{25}(48/625) = 2/25$.

The function Ψ_{25} is continuous, and we have proved that $\Psi_{25}(0) \leq 3/38$ and $\Psi_{25}(48/625) = 2/25$. Its image of $[0, 48/625]$ therefore contains $[3/38, 2/25]$. Together with [Proposition 4.6](#), this proves the proposition.

The exact matrix identities and endpoint comparisons used above are independently checked by `verify_pi2_4_exact.py`. $\square$

4.4 High-density transfers

By [Propositions 4.6](#) and [4.9](#), we have $[0, 2/25] \subseteq \Pi_{2,\infty}^4$. We first use satellite extensions to reach $5/32$, then two further transfers to reach $1/2$, and finally a suspension to cover $[1/2, 1]$. The transversal-link construction realizes the intervening limiting endpoints, while finite rational certificates handle the initial satellite chain.

Proof of [Theorem 4.1](#). We begin with the satellite extension sequence. The forty-two certificate rows in `pi2_4_high_satellite_rows.txt` encode exact applications of [Lemma 4.3](#) extending $[0, 2/25]$ to $[0, 36/251]$. They verify $0 < \rho \leq 1/4$, $S \leq 31/5$, $L \leq 3$, $B \leq M$, and $x_* \leq M < T$, while the proof of [Lemma 4.8](#) gives the common Bernstein-polynomial argument throughout the second parameter range. These rows are checked by `verify_pi2_4_exact.py`.

The remaining satellite extensions form a closed-form family. For an integer $t \geq 8$, use $H = P_3$, with satellite parameter 6 at its center and t at the two leaves, and put $T_t := 5(t-1)/(4(8t-3))$, $B_t := 30t(t-1)/(14t-3)^2$, and $x_t^* := 5(t-1)(2t-3)/(8t-3)^2$. For these parameter choices, the associated parameters $S = 2(8t-3)/(5(t-1))$, $L = 2t/(t-1)$, and $\rho = (2t-3)/(8t-3)$ satisfy $0 < \rho < 1/4$, $S < 31/5$, and $L \leq 16/7 < 3$. They therefore lie in the second parameter range of [Lemma 4.8](#), so the two extension inequalities hold. Starting from $36/251$, this family gives $[0, T_t]$ with $T_t \uparrow 5/32$: the choice $t = 8$ satisfies $B_8 \leq 36/251$ and $x_8^* = 455/3721 \leq 36/251 < T_8$, and for every $t \geq 8$,

$$T_{t-1} - x_t^* = \frac{5(72t^2 - 211t + 114)}{4(8t-11)(8t-3)^2} > 0 \quad \text{and} \quad T_{t-1} - B_t = \frac{5(4t^3 - 20t^2 - 87t - 18)}{4(8t-11)(14t-3)^2} > 0.$$

The first numerator is positive for $t \geq 8$; the second is positive at $t = 8$ and strictly increasing. Therefore the intervals overlap.

The endpoint is exactly realized by the transversal-link construction in [Theorem 3.2](#) with ordinary Turán density $1/2$ and $(a, b) = (8, 5)$, since $\frac{1}{2}(5)_2/8^2 = 5/32$. Hence $[0, 5/32] \subseteq \Pi_{2,\infty}^4$.

All three transfers below admit arbitrary subgraphs of their permitted profiles and hence preserve admissibility and monotonicity. We begin with the *symmetric A^2B^2 transfer*. Let s be a prime power, put $\gamma := 1 - 1/s$, and let $\mathcal{P}_x$ be an admissible monotone class of minimum 2-degree density x . A member G of the resulting class admits a partition $A \dot{\cup} B$ such that $G[A], G[B] \in \mathcal{P}_x$ and every cross edge has profile A^2B^2 . For each $S \in \binom{A}{2}$, require the graph on B with edge set $\{T \in \binom{B}{2} : S \cup T \in G\}$ to be s -colorable; symmetrically, for each $T \in \binom{B}{2}$, require the graph on A with edge set $\{S \in \binom{A}{2} : S \cup T \in G\}$ to be s -colorable. The coloring may depend on S or T , respectively; no common coloring of either part is required. We also allow $s = \infty$, meaning $\gamma = 1$ and imposing no colorability restriction on the cross links.

Claim 4.12. *The resulting class has density $\max\{x, (\gamma + x)/4\}$.*

Proof. Let the part ratios be a, b , with $a + b = 1$. Pairs contained in A and B give the upper bounds $xa^2 + \gamma b^2$ and $\gamma a^2 + xb^2$, respectively. Suppose $a \geq 1/2$ and $x \leq \gamma$. The first expression is smaller. As a convex function of $a \in [1/2, 1]$, it is at most the larger of its endpoint values $(x + \gamma)/4$ and x . The other half is symmetric. If $x > \gamma$, both expressions are at most $x(a^2 + b^2) \leq x$.

For the lower bound, placing all vertices in one part gives x . For the balanced construction with finite s , take asymptotically extremal members of $\mathcal{P}_x$ inside A and B , identify both parts with $\mathbb{F}_s^d$, and declare (aa', bb') to be a cross edge if and only if $\langle a - a', b - b' \rangle \neq 0$. For fixed aa' , the map $b \mapsto \langle a - a', b \rangle$ is an exactly balanced s -coloring, and its link is the corresponding complete s -partite graph; the symmetric statement holds for fixed bb' . Moreover, each pair meeting both A and B belongs to $(s^d - 1)(s^d - s^{d-1}) = (\gamma - o(1))|A||B|$ cross edges. For $s = \infty$, use all cross edges. Thus pairs contained in one part have degree $(x + \gamma)/4 - o(1)$, while pairs meeting both parts have degree $\gamma/2 - o(1)$. If the value from the balanced construction exceeds x , then $x < \gamma/3$, so $\gamma/2 > (x + \gamma)/4$. $\square$

By the claim, if $[0, M] \subseteq \Pi_{2,\infty}^4$, $M \geq \gamma/4$, and $M < \gamma/3$, then the image of $[0, M]$ contains $[\gamma/4, (\gamma + M)/4]$, which overlaps the interval already realized and strictly enlarges it. Repeated application gives right endpoints increasing to $\gamma/3$; hence after finitely many steps the right endpoint exceeds any prescribed real number below $\gamma/3$. The limiting value $\gamma/3$ is realized by the transversal-link construction. Successively take $\gamma = 1/2, 2/3, 3/4, 1$, using the exact identities

$$\frac{1}{2} \frac{(4)_2}{6^2} = \frac{1}{6}, \quad \frac{2}{3} \frac{(4)_2}{6^2} = \frac{2}{9}, \quad \frac{1}{2} \frac{(9)_2}{12^2} = \frac{1}{4}, \quad \frac{1}{2} \frac{(25)_2}{30^2} = \frac{1}{3}.$$

Since $5/32 > 1/8$, these transfers give $[0, 1/3] \subseteq \Pi_{2,\infty}^4$.

The symmetric transfer reaches $1/3$. To continue toward $1/2$, we use ordinary 3-graph classes as links in the A^3B -profile. Fix an admissible monotone class $\mathcal{Q}_q$ of 3-graphs with ordinary Turán density q and an admissible monotone class $\mathcal{P}_t$ of 4-graphs with minimum 2-degree density t . Use a partition $A \dot{\cup} B$ with the following allowed profiles:

- (C1) $G[B] \in \mathcal{P}_t$ and $G[A]$ is empty;
- (C2) all A^2B^2 edges;
- (C3) for each $b \in B$, the 3-graph on A formed by the A^3B -edges containing b belongs to $\mathcal{Q}_q$;
- (C4) no AB^3 edges.

Claim 4.13. *For $0 \leq t \leq 1/2$, put $\zeta_q(t) := q + \sqrt{q^2 + 1 - t}$ and $\Theta_q(t) := (\zeta_q(t)^2 + t)/(1 + \zeta_q(t))^2$. The resulting class has density $\max\{t, \Theta_q(t)\}$.*

Proof. Let $\zeta := a/b$ be the ratio of the part sizes. For the upper bound, average all A^3B links over pairs in A and choose a pair in B of asymptotically minimum degree in $G[B]$. After multiplication by $(1 + \zeta)^2$, the two normalized internal-pair bounds are $1 + 2q\zeta$ and $\zeta^2 + t$, respectively. They meet at $\zeta = \zeta_q(t)$, and the common normalized value is $\Theta_q(t)$. The boundary case $A = \emptyset$ gives t , while $B = \emptyset$ gives zero.

To complete the upper bound, observe that the first normalized bound is decreasing in ζ . The second decreases up to $\zeta = t$ and then increases, while $\zeta_q(t) \geq \sqrt{1 - t} \geq t$ for $t \leq 1/2$. Before the intersection the second bound is smaller, and its maximum on $[0, \zeta_q(t)]$ is $\max\{t, \Theta_q(t)\}$. After the intersection the first bound is smaller and is at most $\Theta_q(t)$. This proves the asserted maximum over the part ratio.

For the lower bound, in each A^3B link take an independent uniformly random relabeling of a member supplied by Lemma 2.2, with edge density and normalized minimum vertex degree at least $q - o(1)$, and take an extremal member of $\mathcal{P}_t$ inside B . Concentration gives the required internal-pair bounds uniformly. An AB -pair has normalized degree $(2\zeta + q\zeta^2)/(1 + \zeta)^2$; at the intersection of the two internal-pair bounds, the numerator of its difference from $\Theta_q(t)$ is $q\zeta^2 + 2(1 - q)\zeta - 1$. For fixed q , this expression is minimized at $t = 1/2$: the quantity $\zeta_q(t)$ decreases with t , whereas $q\zeta^2 + 2(1 - q)\zeta - 1$ increases with ζ for $0 \leq q \leq 1$. At $t = 1/2$, one has $q = (\zeta^2 - 1/2)/(2\zeta)$, and four times the difference numerator is $2\zeta^3 - 4\zeta^2 + 7\zeta - 2$. Its derivative $6\zeta^2 - 8\zeta + 7$ is positive for all real ζ , and at $\zeta = 1/\sqrt{2}$ its value is $4\sqrt{2} - 4 > 0$. Thus AB -pairs are never the bottleneck. $\square$

By the claim, the only value that can improve on t is $\Theta_q(t)$. The function Θ_q is strictly increasing and satisfies $0 < \Theta'_q(t) < 1$ for $0 \leq t \leq 1/2$. With $\zeta = \zeta_q(t)$, differentiation gives $\Theta'_q(t) = (1 + q(\zeta - 1))/((\zeta - q)(\zeta + 1)^3)$. Here $\zeta - q > 0$, while $1 + q(\zeta - 1) = 1 - q + q\zeta > 0$ for $q < 1$ and equals $\zeta > 0$ for $q = 1$; hence $\Theta'_q(t) > 0$. After substituting $q = (\zeta^2 + t - 1)/(2\zeta)$, the positivity of $1 - \Theta'_q(t)$ reduces to $\zeta^4 + 3\zeta^3 - \zeta^2 t + 3\zeta^2 - 3\zeta t + 5\zeta - 4t + 2 > 0$. For fixed $\zeta > 0$, the left side decreases with t . At $t = 1/2$ it is $\zeta^4 + 3\zeta^3 + (5/2)\zeta^2 + (7/2)\zeta > 0$, so the inequality holds throughout $0 \leq t \leq 1/2$. Therefore $\Theta_q(t) - t$ is strictly decreasing.

To decide when the transfer improves the current endpoint, define $p_q(t) := t^3 + (1 + 4q)t^2 + (3 - 4q)t - 1$. With $\zeta = \zeta_q(t)$, direct substitution gives

$$p_q(t) = -\frac{(1 + \zeta)^2(2\zeta t + (1 - t)^2)}{\zeta^2}(\Theta_q(t) - t).$$

The prefactor is negative, so $\Theta_q(t) > t$ if and only if $p_q(t) < 0$.

We next choose a chain of ordinary Turán densities. For integers $m \geq d \geq 2$, the complete m -partite d -graph class consists of all subgraphs of complete m -partite d -graphs, with the partition allowed to vary. This class is admissible and monotone and has ordinary Turán density $(m)_d/m^d$, attained asymptotically by balanced parts. We use $2/9$, $8/27$, $3/8$, $12/25$, $5/9$, $21/32$, $55/72$, and $552/625$. Except for $8/27$, these are the values $(m - 1)(m - 2)/m^2$ realized by the complete multipartite class with $d = 3$. The value $8/27$ is the density of the class of all subgraphs of blowups of the 3-graph with edge set $\{012, 013, 023\}$, because $6x_0(x_1x_2 + x_1x_3 + x_2x_3) \leq 2x_0(1 - x_0)^2 \leq 8/27$.

To pass from a stage with parameter q to the next stage q' , we choose $u \in (1/2, 1)$ so that the q -stage improves endpoints below u^2 and, once u^2 is reached, the image of the q' -stage overlaps the interval already realized. The latter condition $\Theta_{q'}(0) < u^2$ is equivalent to $q' < (2u - 1)/(2u(1 - u))$. The exact overlap certificates are

q	q'	u	$\frac{2u - 1}{2u(1 - u)} - q'$	$-p_q(u^2)$
$2/9$	$8/27$	$7/12$	$44/945$	$23413/995328$
$8/27$	$3/8$	$3/5$	$1/24$	$788/46875$
$3/8$	$12/25$	$8/13$	$3/400$	$92249/4826809$
$12/25$	$5/9$	$7/11$	$17/504$	$769724/44289025$
$5/9$	$21/32$	$13/20$	$9/2912$	$3990373/192000000$
$21/32$	$55/72$	$43/64$	$341/21672$	$2854295/68719476736$
$55/72$	$552/625$	$20/29$	$131/45000$	$825121/594823321$

Every entry in the last two columns is positive. Thus $\Theta_{q'}(0) < u^2$ and $\Theta_q(u^2) > u^2$ for each row. Since $\Theta_q(t) - t$ is decreasing, whenever the current right endpoint t has not reached u^2 , one application of the q -transfer raises it by at least $\Theta_q(u^2) - u^2 > 0$. Hence finitely many applications raise the right endpoint to at least u^2 , after which the q' -stage overlaps the interval already obtained.

The initial image overlaps the interval already realized because $\Theta_{2/9}(0) = ((2 + \sqrt{85})/(11 + \sqrt{85}))^2 < 1/3$, and the first transfer strictly increases the right endpoint because $p_{2/9}(1/3) = -4/81 < 0$. Starting from $[0, 1/3]$, the preceding argument therefore reaches the last stage after finitely many iterations at each value of q . Finally, $p_{552/625}(1/2) = -41/5000 < 0$, so the monotonicity of $\Theta_{552/625}(t) - t$ shows that finitely many iterations of the last stage raise the right endpoint above $1/2$. Thus $[0, 1/2] \subseteq \Pi_{2,\infty}^4$.

The exact endpoint and overlap calculations in this subsection are independently checked by `pi2_4_high_module_certificate.py`.

Having reached $[0, 1/2]$, we use a suspension transfer to cover the larger densities. An analogous suspension map appears in the recent study of single-forbidden degree spectra [1, Theorem 1.1].

Let $\mathcal{P}_x$ be an admissible monotone class of minimum 2-degree density x . The resulting class consists of the 4-graphs admitting a partition $A \dot{\cup} B$ such that $G[A] \in \mathcal{P}_x$, $G[B]$ is empty, every edge of profiles A^3B and A^2B^2 is allowed, and the remaining profiles are forbidden.

To compute the density of this class, let $a := |A|/n$. If both parts have linear size, a pair of asymptotically minimum degree in $G[A]$, a pair in B , and an AB -pair give the upper bounds $1 - (1 - x)a^2$, a^2 , and $2a - a^2$, respectively. The boundary cases $a = 1$ and $a = 0$ give at most x and zero. Since $2a - a^2 \geq a^2$,

the maximum of the minimum of the three bounds occurs where the first two are equal, namely at $a^2 = 1/(2-x)$. Conversely, choose an asymptotically extremal member of $\mathcal{P}_x$ inside A and include all edges of the two cross profiles at this ratio. The three bounds are then attained up to $o(1)$, so the resulting class has density $\tau(x) := 1/(2-x)$. The identity $1/(1-\tau(x)) = 1/(1-x) + 1$ gives $\tau^j([0, 1/2]) = [j/(j+1), (j+1)/(j+2)]$. These intervals cover $[0, 1)$, and the empty forbidden family realizes 1. $\square$

5 Minimum 2-degrees in higher uniformities

The preceding section proves the four-uniform base case. We now lift this interval result to every larger uniformity.

Theorem 5.1. *For every integer $r \geq 5$, one has $\Pi_{2,\infty}^r = [0, 1]$.*

Throughout this section put $p := r - 2$, so $p \geq 3$. Thus a minimum 2-degree is normalized by $\binom{n-2}{p}$. The constructions below use the profile convention from Section 2; in every case arbitrary subgraphs of the allowed edge set are admitted, so the resulting classes are admissible and monotone.

We begin with a two-part construction that transfers densities from the four-uniform case.

Let $\mathcal{P}_x$ be an admissible monotone class of 4-graphs with $\text{dens}_2^4(\mathcal{P}_x) = x$. When $x = 1$, we take $\mathcal{P}_1$ to be the class of all 4-graphs; this class is used only in the construction. Define a class $\mathcal{L}_p(\mathcal{P}_x)$ of $(p+2)$ -graphs. A member admits a partition $V = X \dot{\cup} Y$ and an auxiliary 4-graph $H \in \mathcal{P}_x$ on X . An edge is allowed if it satisfies one of the following conditions.

(F1) It has profile $X^4 Y^{p-2}$, and its four vertices in X form an edge of H .

(F2) It has profile XY^{p+1} .

Define the relevant scaling constants by

$$\kappa_p := \binom{p}{2} \left(\frac{2}{p}\right)^2 \left(\frac{p-2}{p}\right)^{p-2} = \frac{2(p-1)}{p} \left(\frac{p-2}{p}\right)^{p-2}, \quad \text{and} \quad x_p := \frac{(p-2)^2}{2p(p-1)}.$$

Lemma 5.2. *For every $p \geq 3$ and $0 \leq x \leq \min\{1, 3x_p\}$, one has $\text{dens}_2^{p+2}(\mathcal{L}_p(\mathcal{P}_x)) = \kappa_p x$.*

Consequently, $[0, \kappa_p \min\{1, 3x_p\}] \subseteq \Pi_{2,\infty}^{p+2}$. In particular, $[0, ((p-2)/p)^p] \subseteq \Pi_{2,\infty}^{p+2}$ for every $p \geq 3$; when $3x_p \leq 1$, the larger interval $[0, 3((p-2)/p)^p]$ is also contained in the spectrum.

Proof of Lemma 5.2. Let $|X| = (a + o(1))n$, $|Y| = (b + o(1))n$, where $a + b = 1$. Choose a pair $S \in \binom{X}{2}$ of asymptotically minimum 2-degree in the auxiliary 4-graph H . Edges of type (F2) contain only one vertex of X , so they do not contain S . Hence, before and after normalization, respectively,

$$d_G(S) \leq (x + o(1)) \binom{|X| - 2}{2} \binom{|Y|}{p-2} \quad \text{and} \quad \frac{d_G(S)}{\binom{n-2}{p}} \leq x \binom{p}{2} a^2 b^{p-2} + o(1).$$

The right side is maximized at $a = 2/p$ and $b = (p-2)/p$, with maximum $\kappa_p x$. This proves the upper bound when both parts have linear size. If $|X| = o(n)$, a pair in Y has only $o(n^p)$ extensions; if $|Y| = o(n)$, an XX -pair has only $o(n^p)$ extensions. Thus the same upper bound holds when one part has sublinear size.

For the lower bound, use these proportions, choose a near-extremal $H \in \mathcal{P}_x$ on X , and include every edge allowed by this witness. An XX -pair has normalized degree $\kappa_p x - o(1)$. For an XY -pair uv , the edges of type (F2) contribute b^p . Moreover, $3d_H(u) = \sum_{w \in X \setminus \{u\}} d_H(uw) \geq (|X| - 1)\delta_2(H)$, so the edges of type (F1) contribute $x \binom{p}{3} a^3 b^{p-3} = (2/3)\kappa_p x$ asymptotically. Therefore every XY -pair has normalized degree at least $b^p + (2/3)\kappa_p x - o(1)$. Since $b^p = \kappa_p x_p$, this is at least $\kappa_p x - o(1)$ whenever $x \leq 3x_p$.

For a YY -pair, the edges of type (F2) contribute $pab^{p-1} = 2b^{p-1}$. When $p \geq 4$, the edges of type (F1) add $x \binom{p}{4} a^4 b^{p-4}$; for $p = 3$ this term is absent. At the extreme value $x = 3x_p$, direct simplification gives

$$pab^{p-1} + 3x_p \binom{p}{4} a^4 b^{p-4} = 3 \frac{p-1}{p} b^{p-1} > 3b^p = \kappa_p(3x_p)$$

for $p \geq 4$, while for $p = 3$ the contribution from type (F2) is $2/9 > 1/9 = \kappa_3(3x_3)$. The required lower bound for YY -pairs is immediate at $x = 0$, and the difference between their normalized degree and $\kappa_p x$ is affine in x . Hence this bound holds throughout the stated interval and, together with the preceding XX - and XY -pair estimates, proves the matching lower bound. $\square$

The preceding lift yields an interval adjacent to zero. We next construct terminal intervals using ordinary 3-graph links. The source-part organization further develops the part-and-link principle from [25, Section 3.3], now in a symmetric rather than cyclic form: outside transversals carry ordinary 3-graph links, while complete type-2 and transversal profiles protect pairs spanning distinct parts.

Let $\mathcal{Q}_\beta$ be an admissible monotone class of 3-graphs with ordinary Turán density β , and fix an integer $a \geq p + 2$. Partition the vertices into a parts $V_0, \dots, V_{a-1}$. We allow the following profiles.

1. A *type-2 edge* has two vertices in one part V_i , called its source part, and one vertex in each of p distinct other parts.
2. A *type-3 edge* has three vertices in one source part V_i and one vertex in each of $p - 1$ distinct other parts. For every fixed outside transversal, the corresponding 3-graph link inside V_i belongs to $\mathcal{Q}_\beta$.
3. Every transversal edge meeting $p + 2$ distinct parts is allowed.

No other edge profiles are allowed. Call the resulting class $\mathcal{T}_{a,p}(\mathcal{Q}_\beta)$.

Lemma 5.3. *One has*

$$\text{dens}_2^{p+2}(\mathcal{T}_{a,p}(\mathcal{Q}_\beta)) = \lambda_{a,p}(\beta) := \frac{(a-1)_p + \beta p(a-1)_{p-1}}{a^p}.$$

For the endpoint comparisons below, for every integer $q \geq p - 1$ put $U_{q,p} := (q)_p / q^p$ and $W_{q,p} := \binom{p}{2}(q-1)_{p-2} / q^p$. In particular, as β ranges over the ordinary 3-graph terminal interval $[c_3, 1]$, Lemma 5.3 gives $[\lambda_{a,p}(c_3), U_{a,p}] \subseteq \Pi_{2,\infty}^{p+2}$.

For the proof, for a finite index set J , a vector $\mathbf{z} = (z_i)_{i \in J}$, and an integer $d \geq 0$, write

$$e_d(\mathbf{z}) := \sum_{I \in \binom{J}{d}} \prod_{i \in I} z_i, \quad \text{and} \quad e_0(\mathbf{z}) := 1,$$

and use $\mathbf{z}_I$ for the restriction of $\mathbf{z}$ to $I \subseteq J$.

Proof of Lemma 5.3. Consider an arbitrary sequence of members of the class and pass to a subsequence along which $|V_i|/n \rightarrow z_i$ for every $i \in \mathbb{Z}_a$. Write $\mathbf{z} := (z_i)_{i \in \mathbb{Z}_a}$. Put $J := \{i : z_i > 0\}$ and $q := |J|$. Parts outside J contribute only $o(n^p)$ completions. For each $i \in J$, averaging over the pairs in V_i and over all outside transversals yields a pair in V_i whose normalized degree is at most the corresponding expression in the minimum below. We average these expressions over i using the identities

$$\sum_{i \in J} e_p(\mathbf{z}_{J \setminus \{i\}}) = (q-p)e_p(\mathbf{z}_J), \quad \text{and} \quad \sum_{i \in J} z_i e_{p-1}(\mathbf{z}_{J \setminus \{i\}}) = p e_p(\mathbf{z}_J).$$

Maclaurin's inequality then gives

$$\min_{i \in J} \{p! e_p(\mathbf{z}_{J \setminus \{i\}}) + \beta p! z_i e_{p-1}(\mathbf{z}_{J \setminus \{i\}})\} + o(1) \leq \frac{p!}{q} (q-p + \beta p) e_p(\mathbf{z}_J) + o(1) \leq \lambda_{q,p}(\beta) + o(1),$$

with the convention that the last expression is zero when $q < p$. For $q \geq p$, one has $\lambda_{q,p}(\beta) = \prod_{j=1}^{p-1} (1-j/q)(1-p(1-\beta)/q)$. Every factor is nonnegative and nondecreasing in q . Thus $\lambda_{q,p}(\beta) \leq \lambda_{a,p}(\beta)$, which proves the upper bound even when some of the original a parts are empty or sublinear.

For the lower bound take equal parts and use independent uniformly random relabelings of a near-extremal member of $\mathcal{Q}_\beta$ in the type-3 links. For a pair contained in one part, the random type-3 contribution is a sum over $\Theta(n^{p-1})$ independent links. The only genuinely random contribution for a pair meeting two parts occurs when the pair supplies one vertex from the source part, in which case it is a sum over $\Theta(n^{p-2})$ independent links; when the pair supplies no source vertex, the relevant edge count is invariant under relabeling. Since $p \geq 3$, every such sum has $\Omega(n)$ independent terms. Choose an error tending to zero

whose square times n dominates $\log n$. After scaling each summand to $[0, 1]$, Hoeffding's inequality and a union bound over all pairs show simultaneously that every random contribution is at least its expectation minus $o(n^p)$. Thus every pair contained in one part has normalized degree at least $\lambda_{a,p}(\beta) - o(1)$. For a pair lying in two distinct parts, the transversal, type-2, and type-3 profiles give normalized degree

$$\frac{(a-2)_p + \frac{p(p+3)}{2}(a-2)_{p-1} + \beta \frac{p(p-1)(p+4)}{6}(a-2)_{p-2}}{a^p} + o(1).$$

Its surplus over $\lambda_{a,p}(\beta)$ is affine in β . At $\beta = 0$ and $\beta = 1$ it is, respectively,

$$\frac{p(p+1)}{2} \frac{(a-2)_{p-1}}{a^p} \quad \text{and} \quad \frac{p(p-1)(3(a-p) + p-2)}{6} \frac{(a-2)_{p-2}}{a^p}.$$

Both surpluses are positive, so pairs meeting two parts are not bottlenecks. $\square$

We next use four-uniform links to connect consecutive intervals from the preceding construction.

At $\beta = 1$, let $\mathcal{P}_x$ be an admissible monotone class of 4-graphs with $\text{dens}_2^4(\mathcal{P}_x) = x$, and enlarge $\mathcal{T}_{a,p}(\mathcal{Q}_1)$ as follows. For each source part V_i , choose one auxiliary 4-graph $H_i \in \mathcal{P}_x$. Allow a further profile consisting of an edge of H_i together with one vertex in each of $p-2$ distinct other parts. Thus every outside transversal with source V_i uses the same auxiliary 4-graph H_i . Denote the new class by $\mathcal{T}_{a,p}^+(\mathcal{P}_x)$.

Lemma 5.4. *For every $x \in [0, 1]$, one has $\text{dens}_2^{p+2}(\mathcal{T}_{a,p}^+(\mathcal{P}_x)) = U_{a,p} + xW_{a,p}$. Consequently, these classes realize $[U_{a,p}, U_{a,p} + W_{a,p}]$. Moreover, $U_{a,p} + W_{a,p} \geq U_{a+1,p}$.*

Proof. As in the preceding proof, pass to limiting part weights $\mathbf{z} = (z_i)_{i \in \mathbb{Z}_a}$, let J be their positive support, and put $q = |J|$. For a pair contained in V_i , the type-2 and type-3 contributions at $\beta = 1$ are at most $p!e_p(\mathbf{z}_J)$, while the additional four-uniform profile contributes $\frac{p!}{2} x z_i^2 e_{p-2}(\mathbf{z}_{J \setminus \{i\}})$. If $q \leq p-2$, both terms are zero. Otherwise choose $i \in J$ with $z_i \leq 1/q$ and choose a pair of asymptotically minimum degree in the common auxiliary 4-graph H_i . Maclaurin's inequality applied to all q positive weights gives $p!e_p(\mathbf{z}_J) \leq U_{q,p}$. Applied to the other $q-1$ weights, it also gives

$$e_{p-2}(\mathbf{z}_{J \setminus \{i\}}) \leq \binom{q-1}{p-2} \left(\frac{1-z_i}{q-1} \right)^{p-2}.$$

The function $t^2(1-t)^{p-2}$ is increasing for $0 \leq t \leq 2/p$, and $1/q \leq 2/p$ when $q \geq p-1$. Hence the selected pair has normalized degree at most $U_{q,p} + xW_{q,p} + o(1)$.

It remains to compare different support sizes. For $q \geq p$, writing $y = 1/q$ gives

$$U_{q,p} + xW_{q,p} = \prod_{j=1}^{p-2} (1-jy) \left(1 - (p-1)y + x \binom{p}{2} y^2 \right).$$

Every factor in the product is nonincreasing and nonnegative on $y \in (0, 1/p]$. The derivative of the last factor is $(p-1)(xpy - 1) \leq 0$, so the whole expression is nonincreasing in y and hence nondecreasing in q . The remaining comparison from $q = p-1$ to $q = p$ is linear in x . At $x = 0$ it is immediate. At $x = 1$ it is equivalent to $(p/(p-1))^p \leq p+1$, and $(p/(p-1))^p \leq 27/8 \leq p+1$ because $(1-1/p)^p$ is increasing for $p \geq 3$ and equals $8/27$ at $p = 3$. We conclude that $U_{q,p} + xW_{q,p} \leq U_{a,p} + xW_{a,p}$.

This proves the upper bound for every support. Equal parts and near-extremal 4-graphs $H_i \in \mathcal{P}_x$ give normalized degree $U_{a,p} + xW_{a,p} - o(1)$ for pairs contained in one part. At $\beta = 1$, the 3-graph link construction alone exceeds $U_{a,p}$ for every pair meeting two parts by $\frac{p(p-1)(3d+p-2)}{6} \frac{(a-2)_{p-2}}{a^p}$, where $d := a-p \geq 2$. Dividing this surplus by $W_{a,p}$ gives $\frac{(d+1)(3d+p-2)}{3(p+d-1)} \geq 1$. Since $x < 1$, pairs meeting two parts therefore have normalized degree at least $U_{a,p} + xW_{a,p} - o(1)$.

For the overlap, factor $(a-1)_{p-2}/a^{p-2}$ from both sides. The inequality $U_{a,p} + W_{a,p} \geq U_{a+1,p}$ follows from

$$\left(\frac{a}{a+1} \right)^{p-1} \leq 1 - \frac{p-1}{a+1} + \frac{\binom{p-1}{2}}{(a+1)^2} \leq 1 - \frac{p-1}{a} + \frac{\binom{p}{2}}{a^2}.$$

The first inequality follows by truncating the alternating binomial expansion after its quadratic term: the omitted terms alternate with decreasing absolute values because $a \geq p+2$. The second reduces to $a/(a+1) + (p-2)(a/(a+1))^2/2 \leq p/2$. This holds because its left side is at most $1 + (p-2)/2 = p/2$. $\square$

Since $\lambda_{a+1,p}(c_3) < U_{a+1,p} \leq U_{a,p} + W_{a,p}$ and $U_{a,p} \rightarrow 1$, Lemmas 5.3 and 5.4 show that the union of the 3-graph link intervals and the interpolation intervals contains $[\lambda_{p+2,p}(c_3), 1)$.

We now combine the preceding constructions to prove Theorem 5.1.

Proof of Theorem 5.1. The case $r \geq 7$. First suppose that $p = r - 2 \geq 6$. The initial interval from Lemma 5.2 reaches $((p-2)/p)^p$, while the first 3-graph link interval begins strictly below $U_{p+2,p} = (p+1)!/(2(p+2)^{p-1})$ because $c_3 < 1$. To compare these endpoints, set $R_p := 2(p-2)^p(p+2)^{p-1}/(p^p(p+1)!)$. A direct calculation gives $R_6 = 262144/229635 > 1$. Furthermore,

$$\frac{R_{p+1}}{R_p} = \frac{p-1}{p+1} \left(1 + \frac{p^2 + p + 4}{(p-2)(p+1)(p+2)} \right)^p,$$

and hence $\frac{R_{p+1}}{R_p} \geq \frac{p-1}{p+1} \left(1 + \frac{1}{p} \right)^p \geq \frac{2(p-1)}{p+1} > 1$.

The second line uses the bound $1/p$ for the fraction inside the parentheses and the binomial estimate $(1 + 1/p)^p \geq 1 + p(1/p) = 2$. Thus $R_p \geq R_6 > 1$, so the initial and 3-graph link intervals overlap. The interpolation intervals from Lemma 5.4 then extend the realized interval to 1.

It remains to consider $p = 5$. The improved part of Lemma 5.2 gives $[0, 3(3/5)^5] = [0, 729/3125] \subseteq \Pi_{2,\infty}^7$. The first 3-graph link interval has upper endpoint $U_{7,5} = 6!/(2 \cdot 7^4) = 360/2401$. Since $729/3125 - 360/2401 = 625329/7503125 > 0$, the initial interval reaches beyond this endpoint. The interpolation intervals again cover every remaining density below 1. In both cases the empty forbidden family realizes 1.

The case $r = 6$. Now $p = 4$. The improved part of Lemma 5.2 gives $[0, 3/16] \subseteq \Pi_{2,\infty}^6$, while the first 3-graph link interval starts below $5/18$. We cover this gap by a two-part construction.

Let $\mathcal{P}_x$ be an admissible monotone class of 4-graphs with minimum 2-degree density x , and let $\mathcal{Q}_\theta$ be an admissible monotone class of 4-graphs with ordinary Turán density θ . Define a class as follows. A member admits a partition $V = X \dot{\cup} Y$, an auxiliary 4-graph $H \in \mathcal{P}_x$ on X , and a 4-graph $L_S \in \mathcal{Q}_\theta$ on Y for every $S \in \binom{X}{2}$. Its edges may have the following forms:

1. an edge of H together with two vertices of Y ;
2. $S \cup F$, where $S \in \binom{X}{2}$ and $F \in L_S$;
3. one vertex of X together with five vertices of Y .

For $\zeta \in (1/2, 1)$, which will be the limiting ratio $|X|/|Y|$, put $x_\theta(\zeta) := \theta/(3\zeta(1-\zeta))$ and $T_\theta(\zeta) := \theta/((1-\zeta)(1+\zeta)^3)$. This choice makes ζ the relevant local maximum of the internal-pair bound, with value $T_\theta(\zeta)$.

Claim 5.5. *Let $0 < \theta < 1$ and $\zeta \in (1/2, 1)$, and put $x := x_\theta(\zeta)$. Suppose that $x < 1$, $(1-\zeta)(1+\zeta)^3 \leq 1$, and*

$$8\zeta^2\theta - 9\zeta\theta + 3\zeta + 3\theta - 3 \leq 0, \tag{5.1}$$

$$17\zeta^3\theta - 18\zeta^2\theta + 12\zeta^2 + 3\zeta\theta - 12\zeta + 3\theta \leq 0. \tag{5.2}$$

Then the resulting class has density $T_\theta(\zeta)$.

Proof. Consider an arbitrary sequence of members of the resulting class. If $|X| = o(n)$, every pair in Y has degree at most $|X| \binom{|Y|-2}{3} + \binom{|X|}{2} \binom{|Y|-2}{2} + \binom{|X|}{4} = o(n^4)$. If $|Y| = o(n)$, every pair in X has degree at most $\binom{|X|-2}{2} \binom{|Y|}{2} + \binom{|Y|}{4} = o(n^4)$. Thus neither case in which one part has sublinear size gives a positive asymptotic minimum 2-degree.

Suppose now that both parts have linear size, and pass to a subsequence such that $|X|/|Y| \rightarrow s \in (0, \infty)$. Choose a pair of minimum degree in the auxiliary 4-graph H . The definition of $\text{dens}_2^4(\mathcal{P}_x)$ and the ordinary Turán density upper bound for the corresponding link L_S show that the normalized minimum 2-degree is at most $(\theta + 6xs^2)/(1+s)^4 + o(1)$. The derivative of this rational function is $4(3xs(1-s) - \theta)/(1+s)^5$. Since $x = x_\theta(\zeta)$, its stationary points are $1-\zeta$ and ζ ; the first is a local minimum and the second

a local maximum. Its boundary values are θ and 0, while its value at ζ is $T_\theta(\zeta)$. The assumption $(1 - \zeta)(1 + \zeta)^3 \leq 1$ gives $T_\theta(\zeta) \geq \theta$, so $T_\theta(\zeta)$ is the global maximum and gives the required upper bound.

For the lower bound, choose arbitrarily large $H \in \mathcal{P}_x$ on h vertices with $\delta_2(H) \geq (x - o(1))\binom{h-2}{2}$, and take $m = (1/\zeta + o(1))h$ vertices in Y . Choose $L \in \mathcal{Q}_\theta$ on Y with $|L| \geq (\theta - o(1))\binom{m}{4}$, let the 4-graphs L_S be independent uniformly random relabelings of L , and include every allowed edge determined by H and the 4-graphs L_S .

For fixed $u \in X$ and $y \in Y$, the variables $d_{L_{\{u,v\}}}(y)/\binom{m-1}{3}$, indexed by $v \in X \setminus \{u\}$, are independent, lie in $[0, 1]$, and have expectation $\theta - o(1)$. For fixed $\{y, z\} \in \binom{Y}{2}$, the variables $d_{L_S}(\{y, z\})/\binom{m-2}{2}$, indexed by $S \in \binom{X}{2}$, have the same properties. Choosing an error tending to zero whose square times h dominates $\log n$, Hoeffding's inequality and a union bound over the $O(n^2)$ pairs give the corresponding lower bounds simultaneously.

The auxiliary 4-graph H satisfies $3d_H(u) = \sum_{v \in X \setminus \{u\}} d_H(uv)$ and $6|H| = \sum_{S \in \binom{X}{2}} d_H(S)$, so its minimum 2-degree controls every vertex degree and its total number of edges. After multiplication by $(1 + \zeta)^4$, the resulting lower bounds for XX -, XY -, and YY -pairs are, respectively, $\theta + 6x\zeta^2$, $1 + 4\theta\zeta + 4x\zeta^3$, and $4\zeta + 6\theta\zeta^2 + x\zeta^4$. The first normalized bound equals $T_\theta(\zeta)$. After substituting $x = x_\theta(\zeta)$, the other two bounds are at least this value precisely when (5.1) and (5.2), respectively, hold. This proves the matching lower bound. $\square$

We apply the claim on two short parameter ranges. The function T_θ is strictly increasing for $\zeta > 1/2$, since $T'_\theta(\zeta)/T_\theta(\zeta) = 1/(1 - \zeta) - 3/(1 + \zeta) = 2(2\zeta - 1)/(1 - \zeta^2) > 0$. Thus, on any parameter range where the claim applies, the image of that range under T_θ is the interval between its endpoint values. Moreover, $(1 - \zeta)(1 + \zeta)^3$ decreases there and is already $2197/2401 < 1$ at $\zeta = 6/7$. The derivatives of the left sides of (5.1) and (5.2) are $3 + \theta(16\zeta - 9)$ and $24\zeta - 12 + \theta(51\zeta^2 - 36\zeta + 3)$, respectively. Both are positive for $\zeta \geq 6/7$. It is therefore enough, on each of the two parameter intervals below, to check the two lower-bound inequalities at the right endpoint.

It remains to choose the ordinary densities in the two parameter ranges. The values $(4)_4/4^4 = 3/32$ and $(5)_4/5^4 = 24/125$ come from balanced complete multipartite 4-graphs. On both ranges below, $0 < x_\theta(\zeta) < 1$, so the required classes $\mathcal{P}_{x_\theta(\zeta)}$ are available by Theorem 4.1.

First take $\theta = 3/32$ and $6/7 \leq \zeta \leq 18/19$. At the right endpoint, the left sides of (5.1) and (5.2) equal $-33/11552$ and $-46107/219488$. By the claim, this parameter interval realizes every value between $7203/70304$ and $390963/1620896$. Its lower endpoint lies below $3/16$.

Next take $\theta = 24/125$ and $6/7 \leq \zeta \leq 9/10$. At $\zeta = 9/10$, the two left sides equal $-219/6250$ and $-6336/15625$. By the claim, this parameter interval realizes every value in $[57624/274625, 1920/6859]$. The two resulting intervals overlap because $390963/1620896 - 57624/274625 = 13965702771/445138564000 > 0$, and the second overlaps the first 3-graph link interval because $1920/6859 - 5/18 = 265/123462 > 0$.

Since the first 3-graph link interval begins at $\lambda_{6,4}(c_3) = ((5)_4 + 4c_3(5)_3)/6^4 < 360/6^4 = 5/18$, the preceding intervals connect the initial interval to the link intervals. The interpolation intervals from Lemma 5.4 therefore extend the realized interval to 1. Thus the constructions realize $[0, 1]$, and the empty forbidden family realizes 1.

The exact endpoint values and interval comparisons in these two parameter ranges are independently checked by `verify_higher_uniformities_exact.py`.

The case $r = 5$. The case $p = 3$ of Lemma 5.2 gives $[0, 1/9] \subseteq \Pi_{2,\infty}^5$. We next use a two-part construction to reach $1/2$.

For any $x \in [0, 1]$, let $\mathcal{P}_x$ be an admissible monotone class of 4-graphs with minimum 2-degree density x , which is available by Theorem 4.1. Let $\mathcal{Q}_\theta$ and $\mathcal{R}_g$ be admissible monotone classes of 3-graphs and 4-graphs with ordinary Turán densities θ and g , respectively. Define a class of 5-graphs whose members admit a partition $V = X \dot{\cup} Y$ and the following witnesses.

1. There is a 4-graph $H \in \mathcal{P}_x$ on X , and every edge of profile X^4Y has its four vertices in X forming an edge of H .
2. For every $S \in \binom{X}{2}$, the 3-graph $L_S := \{T \in \binom{Y}{3} : S \cup T \in G\}$ belongs to $\mathcal{Q}_\theta$.
3. For every $u \in X$, the 4-graph $R_u := \{Q \in \binom{Y}{4} : \{u\} \cup Q \in G\}$ belongs to $\mathcal{R}_g$.

4. No other profiles are allowed.

The three nonempty profiles are disjoint.

For the density, define the two normalized internal-pair degree functions, for $\zeta \geq 0$, by

$$\varphi_x(\zeta) := \frac{\theta + 3x\zeta^2}{(1 + \zeta)^3}, \quad \text{and} \quad \psi_{\theta,g}(\zeta) := \frac{3g\zeta + 3\theta\zeta^2}{(1 + \zeta)^3}.$$

Their minimum extends continuously to $\zeta \in [0, \infty]$, with value zero at both endpoints. Define $\Phi_{\theta,g}$ and $P_{\theta,g}$ by

$$\begin{aligned} \Phi_{\theta,g}(x) &:= \max_{\zeta \geq 0} \min\{\varphi_x(\zeta), \psi_{\theta,g}(\zeta)\}, \\ P_{\theta,g}(\zeta) &:= 3\theta\zeta^3 + (3g - 9\theta)\zeta^2 + (8\theta - 9g)\zeta + 3g. \end{aligned}$$

Here $\Phi_{\theta,g}$ is the candidate density, while $P_{\theta,g}$ will ensure that mixed pairs are not bottlenecks.

Claim 5.6. *If $P_{\theta,g}(\zeta) \geq 0$ for $0 \leq \zeta \leq 3$, then the resulting class has density $\Phi_{\theta,g}(x)$.*

Proof. We first prove the upper bound. Consider a sequence of members of the resulting class and witnessing partitions. Suppose that both parts have linear size, and pass to a subsequence such that $|X|/|Y| \rightarrow \zeta \in (0, \infty)$. Choose a pair of asymptotically minimum degree in the auxiliary 4-graph H . The only edges containing this pair come from the profile X^4Y determined by H and from its link L_S . After normalization by $\binom{n-2}{3}$, the defining density bounds for $\mathcal{P}_x$ and $\mathcal{Q}_\theta$ therefore give $\varphi_x(\zeta) + o(1)$ as an upper bound for the normalized minimum 2-degree.

For a pair in Y , average over all pairs in $\binom{Y}{2}$. For every $S \in \binom{X}{2}$ and $u \in X$, respectively, one has $\sum_{T \in \binom{Y}{2}} d_{L_S}(T) = 3|L_S|$ and $\sum_{T \in \binom{Y}{2}} d_{R_u}(T) = 6|R_u|$. The ordinary Turán density upper bounds for $\mathcal{Q}_\theta$ and $\mathcal{R}_g$ consequently give a pair in Y of normalized degree at most $\psi_{\theta,g}(\zeta) + o(1)$. Hence the normalized minimum 2-degree is at most $\min\{\varphi_x(\zeta), \psi_{\theta,g}(\zeta)\} + o(1)$.

The boundary cases cannot increase the upper bound obtained from positive limiting part ratios. If $|X| = o(n)$, the same average over pairs in Y , now using the trivial complete-link bounds, gives a pair of degree $O(|X|^2|Y| + |X||Y|^2) = o(n^3)$. If $|Y| = o(n)$, any pair in X has at most $O(|Y||X|^2 + |Y|^3) = o(n^3)$ completions. Maximizing over the limiting ratio proves the upper bound $\Phi_{\theta,g}(x)$.

For the lower bound, fix $\zeta \in (0, \infty)$. Take $|Y|$ along the unbounded sequence supplied by [Lemma 2.2](#) for $\mathcal{R}_g$, and put $|X| = \lfloor \zeta|Y| \rfloor$. On this sequence choose $R \in \mathcal{R}_g$ with $|R| \geq (g - o(1))\binom{|Y|}{4}$ and $\delta_1(R) \geq (g - o(1))\binom{|Y|-1}{3}$. The limit statements in [Section 2](#) also supply $L \in \mathcal{Q}_\theta$ on the same vertex set Y with $|L| \geq (\theta - o(1))\binom{|Y|}{3}$, and $H \in \mathcal{P}_x$ on X with $\delta_2(H) \geq (x - o(1))\binom{|X|-2}{2}$. Take independent uniformly random relabelings of L and R as the links L_S and R_u , respectively, and include every edge of profile X^4Y determined by H .

For an XX -pair, relabeling preserves $|L_S|$, so the lower bound $\varphi_x(\zeta) - o(1)$ is deterministic. A uniformly random relabeling makes each normalized vertex or pair degree have expectation equal to the edge density of the chosen link. For a fixed XY -pair, its 3-link contribution is a sum over $|X| - 1 = \Theta(n)$ independent links, while the minimum vertex-degree bound for R supplies its 4-link contribution. For a fixed YY -pair, the contributions from the 3-links and 4-links are sums over $\binom{|X|}{2} = \Theta(n^2)$ and $|X| = \Theta(n)$ independent variables, respectively. After scaling the summands to $[0, 1]$, choose an error tending to zero whose square times n dominates $\log n$. Hoeffding's inequality and a union bound over the $O(n^2)$ pairs make all these lower bounds hold simultaneously.

The auxiliary 4-graph H also satisfies $3d_H(u) = \sum_{v \in X \setminus \{u\}} d_H(uv) \geq (|X| - 1)\delta_2(H)$. Thus the normalized lower bounds for XX -, XY -, and YY -pairs are, respectively, $\varphi_x(\zeta)$, $(g + 3\theta\zeta + x\zeta^3)/(1 + \zeta)^3$, and $\psi_{\theta,g}(\zeta)$, up to $o(1)$.

It remains to check the XY -pair bound. For $0 \leq \zeta \leq 3$,

$$\min\{\theta + 3x\zeta^2, 3g\zeta + 3\theta\zeta^2\} \leq \frac{\zeta}{3}(\theta + 3x\zeta^2) + \left(1 - \frac{\zeta}{3}\right)(3g\zeta + 3\theta\zeta^2),$$

and three times the difference between the XY -pair numerator and the right side is $P_{\theta,g}(\zeta)$. For $\zeta \geq 3$, the difference between the XY -pair numerator and $\theta + 3x\zeta^2$ is $g - \theta + 3\theta\zeta + x\zeta^2(\zeta - 3) \geq 0$. The hypothesis on

$P_{\theta,g}$ therefore makes the degree of every XY -pair no smaller than the minimum of the two internal-pair bounds. Maximizing over ζ gives the matching lower bound for the density of the resulting class. $\square$

We use the four parameter pairs $(8/27, 3/32)$, $(12/25, 3/32)$, $(240/289, 3/32)$, and $(1, 120/343)$ to form an overlapping chain from $1/9$ to $1/2$. All are explicit ordinary Turán densities: $8/27$ comes from the class of all subgraphs of blowups of the 3-graph with edge set $\{012, 013, 023\}$, while the others come from balanced complete multipartite hypergraphs.

To verify the hypothesis of the claim exactly, the degree-three Bernstein coefficients of $P_{\theta,g}$ on the indicated subintervals are as follows:

(θ, g)	$[u, v]$	Bernstein coefficients
$(8/27, 3/32)$	$[0, 3/2]$	$9/32, 1805/1728, 65/3456, 235/1152$
	$(3/2, 3)$	$235/1152, 1345/3456, 3085/1728, 2129/288$
$(12/25, 3/32)$	$[0, 3/2]$	$9/32, 2847/1600, 159/640, 351/640$
	$(3/2, 3)$	$351/640, 543/640, 4767/1600, 9441/800$
$(240/289, 3/32)$	$[0, 3/2]$	$9/32, 58839/18496, 25395/36992, 44595/36992$
	$(3/2, 3)$	$44595/36992, 63795/36992, 97239/18496, 186921/9248$
$(1, 120/343)$	$[0, 3/4]$	$360/343, 776/343, 10891/5488, 32523/21952$
	$(3/4, 3/2)$	$32523/21952, 10741/10976, 1375/5488, 1545/2744$
	$(3/2, 3)$	$1545/2744, 815/686, 4099/686, 8592/343$

Every coefficient is positive. Thus, by the claim, the density of the resulting class for each of the four parameter pairs is $\Phi_{\theta,g}(x)$.

We claim that, for the four parameter pairs above, the images of $[0, 1)$ under $\Phi_{\theta,g}$ contain $[1/9, 3/20]$, $[3/20, 47/200]$, $[47/200, 3/8]$, and $[3/8, 1/2]$, respectively. A completely rational endpoint certificate is obtained from the pairs

(θ, g)	ζ_-	ζ_+
$(8/27, 3/32)$	$2/5$	$3/2$
$(12/25, 3/32)$	$12/25$	$5/3$
$(240/289, 3/32)$	$89/170$	$9/5$
$(1, 120/343)$	$2/5$	$41/43$

For each row, write $[L, R]$ for the corresponding target interval above. At ζ_- the value at $x = 0$ is strictly below L , while at ζ_+ the value at $x = 1$ is strictly above R . The corresponding positive gaps are

	$\theta - (3g\zeta_- + 3\theta\zeta_-^2)$	$L - \varphi_0(\zeta_-)$	$\min\{\varphi_1, \psi_{\theta,g}\}(\zeta_+) - R$
1	$449/10800$	$29/9261$	$1/200$
2	$1653/125000$	$1959/1013060$	$269/409600$
3	$9903/26726720$	$577013/3474795800$	$400233/25376512$
4	$859/8575$	$29/2744$	$1/148176$

To justify that these local evaluations control the global maximum, make the change of variable $u = \zeta/(1 + \zeta)$, which identifies $[0, \infty]$ with $[0, 1]$. In the variables x and u , the max-min objective defining $\Phi_{\theta,g}$ extends continuously to $[0, 1] \times [0, 1]$. Its maximum over u , namely $\Phi_{\theta,g}(x)$, is therefore continuous on $[0, 1]$.

To bound $\Phi_{\theta,g}(0)$, note that $\varphi_0(\zeta) = \theta/(1 + \zeta)^3$ is decreasing, whereas $\psi'_{\theta,g}(\zeta) = 3(g + 2(\theta - g)\zeta - \theta\zeta^2)/(1 + \zeta)^4$. For each row, the quadratic in parentheses is positive on $[0, \zeta_-]$. Indeed, it is concave, is positive at zero, and its values at the four right endpoints are, respectively, $4501/21600$, $176979/500000$, $425961/668168$, and $6088/8575$. The first two gaps in the table therefore imply $\Phi_{\theta,g}(0) < L$: for $\zeta \leq \zeta_-$ the minimum is at most $\psi_{\theta,g}(\zeta_-) < \varphi_0(\zeta_-) < L$, while for $\zeta \geq \zeta_-$ it is at most $\varphi_0(\zeta) \leq \varphi_0(\zeta_-) < L$.

At $x = 1$, the final gap gives $\Phi_{\theta,g}(1) > R$. By strictness and continuity, some $x_+ < 1$ still satisfies $\Phi_{\theta,g}(x_+) > R$. The intermediate value theorem shows that the realized image for $0 \leq x \leq x_+$ contains $[L, R]$. Applying this intermediate-value argument to the four overlapping rows yields $[0, 1/2] \subseteq \Pi_{2,\infty}^5$.

The displayed Bernstein coefficients and endpoint gaps are independently checked by `verify_higher_uniformities_exact.py`.

It remains to extend $[0, 1/2]$ to all densities below one. We again use a suspension transfer. Let $\mathcal{P}_x$ be an admissible monotone class of 5-graphs with minimum 2-degree density x . The resulting class consists of the 5-graphs admitting a partition $V = A \dot{\cup} B$ such that $G[A] \in \mathcal{P}_x$, $G[B]$ is empty, every edge of profiles A^4B and A^3B^2 is allowed, and the remaining profiles are forbidden.

To compute the density of this class, put $a := |A|/n$. If both parts have linear size, a pair of asymptotically minimum degree in $G[A]$ and a pair in B give the upper bounds $\mu_x(a) := xa^3 + 3a(1-a)$ and $\nu(a) := a^3$, respectively. The corresponding bound for an AB -pair is $a^2(3-2a)$, which is at least $\nu(a)$ for $0 \leq a \leq 1$ and therefore does not determine the minimum. The boundary cases $a = 1$ and $a = 0$ give at most x and zero. Conversely, an asymptotically extremal member of $\mathcal{P}_x$ inside A , together with all edges of the two cross profiles, attains these three bounds up to $o(1)$. Thus the class density is the maximum over $a \in [0, 1]$ of $\min\{\mu_x(a), \nu(a)\}$. For $0 \leq x < 1$, the unique positive intersection of the two functions satisfies $(1-x)a^2 = 3(1-a)$.

Solving this equation, put $a_x := 6/(3 + \sqrt{21-12x})$ and $f(x) := a_x^3$. On $[0, a_x]$, the smaller function is the increasing function ν . On $[a_x, 1]$, the smaller function is μ_x . Moreover, $\mu'_x(a)/3 = xa^2 + 1 - 2a$. This convex quadratic is nonpositive at both endpoints of $[a_x, 1]$, because its values there are $(a_x + 2)(a_x - 1)$ and $x - 1$. Hence μ_x is nonincreasing on that interval, and the resulting class has density $f(x)$.

The function f is continuous and strictly increasing, with $f(0) = 6\sqrt{21} - 27 < 1/2$. For $x < 1$, the intersection equation gives $a_x^2(f(x) - x) = (a_x - 1)^2(a_x^3 + 2a_x^2 + 3a_x + 3) > 0$.

Put $M_0 := 1/2$ and $M_{j+1} := f(M_j)$. If $[0, M_j]$ is realized, applying the suspension to classes with densities $x \in [0, M_j]$ realizes $[f(0), M_{j+1}]$. Since $f(0) < M_j$, the union is $[0, M_{j+1}]$; induction therefore gives $[0, M_j] \subseteq \Pi_{2,\infty}^5$ for every j . The sequence (M_j) is strictly increasing and bounded above by 1. Its limit is a fixed point of the continuous function f ; the inequality $f(x) > x$ for $x < 1$ forces that limit to be 1. Hence every value below 1 is realized, and the empty forbidden family realizes 1. $\square$

6 Exact computational certificates

The density formulas and the estimates valid throughout the relevant parameter ranges are proved in Sections 4.1 to 4.4. In particular, Lemma 4.3 reduces each finite satellite extension to a certificate row of the kind described in Section 4.2. Such a row specifies the construction data, a previously realized endpoint M , and the resulting target T ; the derived values S, K, L, ρ, B , and x_* verify the applicable analytic parameter range and the inequalities $B \leq M$ and $x_* \leq M < T$. Thus one row certifies the extension from $[0, M]$ to $[0, T]$.

The three data files below store the certificate rows omitted from the body of the paper. The certificate folder is available at https://xliu2022.github.io/CodegreeTuranInterval_certificates/. The verification scripts recompute the recorded quantities and check that consecutive rows form the claimed interval chains. All calculations use integers or Python's `Fraction` class, so no floating-point computation enters the proof.

The certificates for the four-uniform case are contained in the following three supplementary files.

- (i) `weighted_path_chain_certificate.txt` contains 113 homogeneous weighted-path records that extend the realized interval from $1/114$ to $1/20$.
- (ii) `heterogeneous_satellite_chain_certificate.txt` contains 57 heterogeneous records in three low-density batches, the last of which ends at $3996/50315$.
- (iii) `pi2_4_high_satellite_rows.txt` contains 42 high-density satellite records that extend the realized interval from $2/25$ to $36/251$.

In the homogeneous file, the auxiliary graph is specified by its number of path components. The heterogeneous file specifies an allowed isomorphism class or path-forest family, from which the verifier reconstructs a labeled witness. The high-density file gives the auxiliary graph by an explicit edge list on $\{0, \dots, m-1\}$. In these files, `None` denotes $s_i = \infty$.

The three verification scripts have the following roles.

- (i) The verifier `verify_pi2_4_exact.py` checks every record in the three files above. For each heterogeneous record, it enumerates every labeled graph in the specified isomorphism class or path-forest

family and requires an exact witness for the recorded parameters. The same verifier checks the D_{25} support-coverage identity, $\det \mathbf{C} = 4$, the entries of $\mathbf{C}^{-1}$ and $\mathbf{K}_0$, the three principal blocks, and the endpoint gaps used in the D_{25} extension.

- (ii) The optional script `pi2_4_high_module_certificate.py` gives an independent symbolic cross-check of the high-density module. It rechecks the 42 satellite records and verifies the analytic-box identities and monotonicity bounds, the closed-form P_3 tail, the ordinary-link overlaps and mixed-pair bound, and the transversal-link endpoints. This cross-check requires SymPy; the two main verifiers use only Python's standard library.
- (iii) The verifier `verify_higher_uniformities_exact.py` checks the remaining finite calculations in the higher-uniformity proof. These are the Bernstein coefficients and endpoint gaps for the four parameter choices used when $r = 5$, and the safety values, image endpoints, and interval overlaps for the two parameter ranges used when $r = 6$.

To reproduce the checks, use `python3` to run all three scripts from the manuscript directory; the scripts are in `CodegreeTuranInterval_certificates`. Each script stops with an error if any check fails, and the expected output and SHA-256 hashes are recorded in the README file in that directory. The scripts verify only the finite identities, substitutions, and inequalities described above; they do not replace the analytic arguments in the preceding sections.

Acknowledgments

H.L. was supported by the National Natural Science Foundation of China (12501487), by the China Scholarship Council, and by the Institute for Basic Science (IBS-R029-C4). X.L. was supported by the Excellent Young Talents Program (Overseas) of the National Natural Science Foundation of China.

Declaration on the use of AI

The authors used generative AI tools to discuss proof organization, test local calculations, check proofs, and improve exposition.

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